

INTERNATIONAL STANDARD



**Cable networks for television signals, sound signals and interactive services –
Part 13-1: Bandwidth expansion for broadcast signal over FTTH system**



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INTERNATIONAL STANDARD



**Cable networks for television signals, sound signals and interactive services –
Part 13-1: Bandwidth expansion for broadcast signal over FTTH system**

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INTERNATIONAL ELECTROTECHNICAL COMMISSION

**CABLE NETWORKS FOR TELEVISION SIGNALS,
SOUND SIGNALS AND INTERACTIVE SERVICES –****Part 13-1: Bandwidth expansion for broadcast signal over FTTH system**

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International Standard IEC 60728-13-1 has been prepared by technical area 5: Cable networks for television signals, sound signals and interactive services, of IEC technical committee 100: Audio, video and multimedia systems and equipment.

This second edition cancels and replaces the first edition published in 2012. This edition constitutes a technical revision.

This edition includes the following significant technical changes with respect to the previous edition.

- Transmission frequency was expanded in order to achieve satellite signal for 4 K video service. The transmission frequency over FTTH would be 3 300 MHz.
- High signal modulation case like 16 APSK and 32 APSK was added in order to correspond to transmission for 4 K video service.

The text of this International Standard is based on the following documents:

FDIS	Report on voting
100/2927/FDIS	100/2959/RVD

Full information on the voting for the approval of this International Standard can be found in the report on voting indicated in the above table.

This document has been drafted in accordance with the ISO/IEC Directives, Part 2.

A list of all parts in the IEC 60728 series, published under the general title *Cable networks for television signals, sound signals and interactive services*, can be found on the IEC website.

The committee has decided that the contents of this document will remain unchanged until the stability date indicated on the IEC website under "<http://webstore.iec.ch>" in the data related to the specific document. At this date, the document will be

- reconfirmed,
- withdrawn,
- replaced by a revised edition, or
- amended.

A bilingual version of this publication may be issued at a later date.

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INTRODUCTION

Standards and deliverables of the IEC 60728 series deal with cable networks including equipment and associated methods of measurement for headend reception, processing and distribution of television and sound signals and for processing, interfacing and transmitting all kinds of data signals for interactive services using all applicable transmission media. These signals are typically transmitted in networks by frequency-multiplexing techniques.

This includes for instance

- regional and local broadband cable networks,
- extended satellite and terrestrial television distribution systems,
- individual satellite and terrestrial television receiving systems,

and all kinds of equipment, systems and installations used in such cable networks, distribution and receiving systems.

The extent of this standardization work is from the antennas and/or special interfaces to the headend or other interface points to the network up to any terminal interface of the customer premises equipment.

The standardization work will consider coexistence with users of the RF spectrum in wired and wireless transmission systems.

The standardization of any user terminals (i.e. tuners, receivers, decoders, multimedia terminals, etc.) as well as of any coaxial, balanced and optical cables and accessories thereof is excluded.

CABLE NETWORKS FOR TELEVISION SIGNALS, SOUND SIGNALS AND INTERACTIVE SERVICES –

Part 13-1: Bandwidth expansion for broadcast signal over FTTH system

1 Scope

The purpose of this part of IEC 60728 is the precise description of an FTTH (fibre to the home) system for expanding broadband broadcast signal transmission from CATV services only, towards CATV plus broadcast satellite (BS) plus communication satellite (CS) services, additionally to other various signals such as data services.

The scope is limited to the RF signal transmission over FTTH systems.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60068-1:2013, *Environmental testing – Part 1: General and guidance*

IEC 60728-1:2014, *Cable networks for television signals, sound signals and interactive services – Part 1: System performance of forward paths*

IEC 60728-6:2011, *Cable networks for television signals, sound signals and interactive services – Part 6: Optical equipment*

IEC 60728-13:2010, *Cable networks for television signals, sound signals and interactive services – Part 13: Optical systems for broadcast signal transmissions*

IEC 60728-113:—, *Cable networks for television signals, sound signals and interactive services – Part 13: Optical systems for broadcast signal transmissions*¹

IEC 61280-1-3:2010, *Fibre optic communication subsystem test procedures – Part 1-3: General communication subsystems – Central wavelength and spectral width measurement*

ITU-T Recommendation G.694.1, *Spectral grids for WDM applications: DWDM frequency grid*

ITU-T Recommendation G.694.2, *Spectral grids for WDM applications: CWDM wavelength grid*

3 Terms, definitions, symbols and abbreviated terms

3.1 Terms and definitions

For the purposes of this document, the following terms and definitions apply.

¹ Under preparation. Stage at the time of publication: IEC ACDV 60728-113: 2017.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- IEC Electropedia: available at <http://www.electropedia.org>
- ISO Online browsing platform: available at <http://www.iso.org/obp>

3.1.1

regional broadband cable network

network designed to provide sound and television signals as well as signals for interactive services to a regional area covering several towns and/or villages

3.1.2

local broadband cable network

network designed to provide sound and television signals as well as signals for interactive services to a local area (e.g. one town or one village)

3.1.3

extended satellite television distribution network or system

distribution network or system designed to provide sound and television signals received by satellite receiving antenna to households in one or more buildings

Note 1 to entry: This kind of network or system can be combined with terrestrial antennas for the additional reception of TV and/or radio signals via terrestrial networks.

Note 2 to entry: This kind of network or system can also carry control signals for satellite switched systems or other signals for special transmission systems (e.g. MoCA or WiFi) in the return path direction.

3.1.4

extended terrestrial television distribution network or system

distribution network or system designed to provide sound and television signals received by terrestrial receiving antennas to households in one or more buildings

Note 1 to entry: This kind of network or system can be combined with a satellite antenna for the additional reception of TV and/or radio signals via satellite networks.

Note 2 to entry: This kind of network or system can also carry other signals for special transmission systems (e.g. MoCA or WiFi) in the return path direction.

3.1.5

individual satellite television receiving system

system designed to provide sound and television signals received from satellite(s) to an individual household

Note 1 to entry: This kind of system can also carry control signals for satellite switched systems or other signals for special transmission systems (e.g. MoCA or WiFi) in the return path direction.

3.1.6

individual terrestrial television receiving system

system designed to provide sound and television signals received via terrestrial broadcast networks to an individual household

Note 1 to entry: This kind of system can also carry other signals for special transmission systems (e.g. MoCA or WiFi) in the return path direction.

3.1.7

optical transmitting unit

optical transmitter

transmit fibre optic terminal device accepting at its input port an electrical signal and providing at its output port an optical carrier modulated by that input signal

Note 1 to entry: For the purposes of this document, optical transmitters may have more than one input port accepting electrical RF signals.

Note 2 to entry: This piece of equipment amplifies frequency multiplexed electrical signals and converts these electrical signals into optical signals. The optical wavelength is a 1 500 nm band (1 550 nm $\pm$ 10 nm in 1 530 nm to 1 625 nm region).

Note 3 to entry: The wavelength and necessary wavelength separation are described in Annexes B and C, respectively.

[SOURCE: IEC 60728-13:2010, 3.1.1, modified — Note 3 has been added.]

3.1.8

optical receiving unit

optical receiver

receive fibre optic terminal device accepting at its input port a modulated optical carrier, and providing at its output port the corresponding demodulated electrical signal (with the associated clock, if digital)

Note 1 to entry: For the purposes of this document, optical receivers may have more than one output port providing electrical RF signals.

[SOURCE: IEC TR 61931:1998, 2.9.7, modified — Note 1 has been added.]

3.1.9

optical amplifier

optical waveguide device containing a suitably pumped, active medium which is able to amplify an optical signal

Note 1 to entry: In this document, erbium doped fibre amplifier (EDFA) is used for amplification in the 1 550 nm band.

Note 2 to entry: There are several methods which several wavelengths are amplified. The term "erbium doped fibre amplifier (EDFA)" is the synonym of optical amplifier in this document.

[SOURCE: IEC TR 61931, 2.7.75, modified — Notes 1 and 2 have been added.]

3.1.10

fibre optic branching device

optical fibre coupler

splitter

optical fibre device, possessing three or more optical ports, which shares optical power among its ports in a predetermined fashion, at the same wavelength or wavelengths, without wavelength conversion

Note 1 to entry: The ports may be connected to fibres, detectors, etc.

[SOURCE: IEC TR 61931:1998, 2.6.21, modified — The term "splitter" has been added, "optical fibre branching" device has been deleted and "optical fibre coupler" is a preferred rather than a deprecated term.]

3.1.11

multiplexing device

WDM device

wavelength selective branching device (used in WDM transmission systems) in which optical signals can be transferred between two predetermined ports, depending on the wavelength of the signal

[SOURCE: IEC TR 61931:1998, 2.6.51]

3.1.12

optical modulation index

optical modulation index of k^{th} RF signal, OMI_k , which is defined as

$$OMI_k = \frac{\phi_h - \phi_l}{\phi_h + \phi_l}$$

where

ϕ_h is the highest, and

ϕ_l is the lowest instantaneous optical power of the intensity modulated optical signal, and

k is the considered RF signal

Note 1 to entry: This definition does not apply to systems where the input signals are converted and transported as digital baseband signals. In this case, the terms modulation depth or extinction ratio defined in 2.6.79 and 2.7.46 of IEC TR 61931:1998 are used. A test procedure for extinction ratio is described in IEC 61280-2-2.

[SOURCE: IEC 60728-6:2011, 3.1.10, modified — The definition has been clarified and Notes 1 and 2 have been replaced by a new Note 1.]

3.1.13

total optical modulation index

resulting optical modulation index when more than one RF signal is transmitted, OMI_{tot} , which is defined as

$$OMI_{tot} = \sqrt{\sum_{k=1}^K OMI_k^2}$$

where

OMI_k is the optical modulation index of the k -th RF signal;

K is the total number of RF signals.

3.1.14

relative intensity noise

RIN

ratio of the mean square of the intensity fluctuations in the optical power of a light source to the square of the mean of the optical output power

Note 1 to entry: *RIN* is usually expressed in dB (Hz⁻¹) resulting in negative values.

Note 2 to entry: The value of *RIN* can also be calculated from the results of a signal-to-noise measurement for the system.

[SOURCE: IEC 60728-13:2010, definition 3.1.8]

3.1.15

responsivity

ratio of an optical detector's electrical output to its optical input at a given wavelength

Note 1 to entry: The responsivity is generally expressed in ampere per watt or volt per watt of incident radiant power.

Note 2 to entry: Sensitivity is sometimes used as an imprecise synonym for responsivity.

Note 3 to entry: The wavelength interval around the given wavelength may be specified.

[SOURCE: IEC 60728-6, 3.1.14]

3.1.16

wavelength

distance covered in a period by the wavefront of a harmonic plane wave

Note 1 to entry: The wavelength λ of light in vacuum is given by

$$\lambda = \frac{c}{f}$$

where

c is the speed of light in vacuum ($c = 2,997\,92 \times 10^8$ m/s);

f is the optical frequency.

Although the wavelength in dielectric material, such as fibres, is shorter than in vacuum, only the wavelength of light in vacuum is used.

[SOURCE: IEC 60728-13:2010, 3.1.10]

3.1.17

central wavelength

average of those wavelengths at which the amplitude of a light source reaches or last falls to half of the maximum amplitude

[SOURCE: IEC 60728-6:2011, 3.1.26, modified — The term "centre wavelength" has been replaced by "central wavelength".]

3.1.18

quadrature amplitude modulation

QAM

amplitude modulation by two separate signals of two sinusoidal carriers having the same amplitude and frequency but being in phase quadrature, the modulated signals being added for transmission in a single channel

[SOURCE: IEC 60050-702:1992, 702-06-63]

3.1.19

QAM signal

signal utilizing quadrature amplitude modulation as its modulation method

3.1.20

orthogonal frequency division multiplexing

OFDM

orthogonal frequency division multiplexing is one of the multiplexing schemes used for the transportation of terrestrial digital broadcasting SDTV and HDTV signals

Note 1 to entry: OFDM is based on the idea of frequency-division multiplexing, where each frequency channel is modulated with a simpler modulation, and the frequencies and modulation of FDM are arranged to be orthogonal with each other, which almost eliminates the interference between channels.

[SOURCE: IEC 60728-13:2010, 3.1.14, modified — OFDM has been specified and "OFDM signal" has been deleted and given as a new entry.]

3.1.21

OFDM signal

signal utilizing orthogonal frequency division multiplexing as the multiplexing scheme

3.1.22

phase shift keying

PSK

angle modulation in which each significant condition in a modulating discretely-timed signal is represented by a specified phase of a periodic sinusoidal oscillation.

[SOURCE: IEC 60050-721:1991, 721-06-07, modified — The preferred terms "phase shift modulation" and "phase shift signalling" have been deleted.]

3.1.23**PSK signal**

signal utilizing phase shift keying as its modulation method

3.1.24**amplitude phase shift keying****APSK**

digital modulation scheme that conveys data by changing, or modulating, both the amplitude and the phase of a reference signal

Note 1 to entry: APSK combines both amplitude-shift keying (ASK) and phase-shift keying (PSK) to increase the symbol-set.

3.1.25**APSK signal**

signal utilizing amplitude phase shift keying as its modulation scheme

3.1.26**level of digitally modulated signal**

RMS power of the signal within the channel bandwidth (S)

Note 1 to entry: The level of digital signal is the average electrical power of the overall signal comprised of each signal and is not the individual signal level of the multi-carrier signal, as shown in Table 1.

Table 1 – Level of RF signals

Signal		Level detection	Symbol	Remarks
Digitally modulated signals	QAM signal	RMS value	S	The value is averaged over a sufficiently long period of time compared to the period of the lowest frequency used for the modulation.
	OFDM signal			
	xPSK signal ^a			
	16 APSK signal			
	32 APSK signal			
FM audio carrier ^b		RMS value	C_{rms}	The carrier level is a constant value.
^a xPSK means QPSK, 8PSK, TC8PSK, etc.				
^b FM radio is not a kind of digitally modulated signals. However, it may exist in some digitally modulated optical broadcast system.				

Note 2 to entry: The level of digitally modulated signal can be expressed in dB(mW) or in dB(μV) referred to 75 Ω.

3.1.27 **S/N ratio**

signal to noise ratio for a digitally modulated signal in the RF band

Note 1 to entry: In this document only digitally modulated carriers are considered. S/N is used only for a digitally modulated signal which expresses the same as $S_{D,RF}/N$ used in IEC 60728-1:2014, 3.1.80.

3.1.28 **D/U ratio**

ratio of desired signal level, D , to undesired signal level, U

Note 1 to entry: The D/U ratio is generally used for multiple frequency interference as CSO and CTB, for single frequency interference as SCR .

Note 2 to entry: D/U ratio is expressed in dB.

[SOURCE: IEC 60728-13:2010, definition 3.1.20, modified — CCR in note 1 has been changed to SCR .]

3.1.29

single or multiple frequency interference ratio

ratio of desired signal level to undesired signal level

Note 1 to entry: The ratio of desired signal level, D (dB(μV)), to undesired signal level, U (dB(μV)) is given by

$$D/U \text{ (dB)} = D - U$$

Note 2 to entry: The desired and the undesired signals can also be expressed both in dB(mW).

Note 3 to entry: Note the similarity of the definition to the definition of S/N ratio.

3.1.30

optical line terminal

OLT

central office-terminal equipment that is linked with the Optical Network Unit (ONU) in the customer premises

Note 1 to entry: The OLT usually connects with headend equipment.

[SOURCE: IEC 60728-13:2010, 3.1.22]

3.1.31

optical network unit

ONU

device at the customer premises equipment that is linked with the Optical Line Terminal (OLT)

Note 1 to entry: The ONU usually connects with customer premises equipment.

3.1.32

video-optical network unit

V-ONU

terminal unit that changes the optical signal of a broadcast system into an electric signal

Note 1 to entry: The term V-ONU is used as the synonym of optical receiver (O/E) in this document.

[SOURCE: IEC 60728-13:2010, 3.1.24]

3.1.33

stimulated Brillouin scattering

SBS

non-linear scattering of optical radiation characterized by a frequency shift as for the Raman scattering, but accompanied by a lower frequency (acoustical) vibration of the medium lattice; the light is scattered backward with respect to the incident radiation

Note 1 to entry: In silica fibres the frequency shift is typically around 10 GHz.

Note 2 to entry: SBS results in loss of optical level and affects the performance of analogue optical system.

Note 3 to entry: The frequency shift is characterized by a frequency downshift (that is to a longer wavelength) due to a GHz frequency acoustical vibration (frequency downshift is 10 GHz or 11 GHz, and gain bandwidth 20 MHz).

[SOURCE: IEC 60728-13:2010, 3.1.25]

3.1.34

stimulated Raman scattering

SRS

non-linear scattering of optical radiation characterized by a wavelength shift and accompanied by very high frequency vibration of the medium lattice, strongly enhanced by the presence of already scattered radiation

Note 1 to entry: In silica fibres the wavelength shift is typically around 100 nm for an exciting radiation with a wavelength around 1 550 nm.

Note 2 to entry: Stimulated Raman scattering can occur in both forward and backward directions and can cause crosstalk between optical signals of different wavelengths.

Note 3 to entry: Frequency downshift is about 13 THz and gain bandwidth about 20 GHz.

[SOURCE: IEC 60728-13:2010, 3.1.27]

3.1.35

cross-phase modulation

XPM

modulation caused by the nonlinear refractive index of the fibre material

Note 1 to entry: It has a relationship with the wavelength spacing in optical transmission system. The more spacing becomes broader, the more XPM value decreases. In such WDM system having 1 490 nm (communication signal) and 1 550 nm (broadcast signal) wavelengths, XPM becomes negligibly small compared with SRS due to this relationship.

Note 2 to entry: XPM affects the performance of the wavelength division multiplex system.

[SOURCE: IEC 60728-13:2010, 3.1.28, modified — The definition has been improved.]

3.1.36

signal to crosstalk ratio

SCR

level difference of CATV broadcast signal level and worst case level of another service's single frequency crosstalk signal measured at an RF output port of an optical receiver for CATV broadcast service

$$SCR = D_{CATV} - U_{OtherService}$$

where

D_{CATV} is the nominal level of a CATV broadcast signal in dB(μV) at an RF output port of an optical CATV broadcast receiver

$U_{OtherService}$ is the worst case level of another service's single frequency crosstalk in dB(μV), within the measurement signal bandwidth, at an RF output port of an optical CATV broadcast receiver. The value of $U_{OtherService}$ is mainly due to the Raman scattering effect from other services such as xPON.

Note 1 to entry: *SCR* is expressed in dB.

3.2 Symbols

The following graphical symbols are used in the figures of this document. These symbols are either listed in IEC 60617 or based on symbols defined in IEC 60617.

	Optical transmitter based on [IEC 60617-S00213 (2001-07)]		Optical amplifier based on [IEC 60617-S01239 (2001-07)]
	Optical fibre [IEC 60617-S01318 (2001-07)]		Variable attenuator [IEC 60617-S01245, modified (2001-07)]
	Optical receiver based on [IEC 60617-S00213 (2001-07)]		Power meter [IEC 60617-S00059, IEC 60617-S00910 (2001-07)]
	Electrical spectrum analyser based on [IEC 60617-S00059, IEC 60617-S00910 (2001-07)]		Amplifier [IEC 60617-S01239 (2001-07)]
	Ammeter based on [IEC 60617-S00059, IEC 60617-S00910 (2001-07)]		Photodiode with fibre pigtail [IEC 60617-S01327 (2001-07)]
	coupler [IEC 60617-S00059, IEC 60617-S01188 (2001-07)]		Optical filter
	Optical terminator [IEC 60617-S01389, IEC 60617-S01318 (2001-07)]		NF meter
	Optical splitter based on [IEC TR 61930:1998, 3.33.1]		Television set
	Video optical network unit		WDM filter

3.3 Abbreviated terms

AGC automatic gain control
APSK amplitude phase shift keying

AM amplitude modulation
APC angled physical contact optical connector

ATC	automatic temperature control	BER	bit error ratio
BS	broadcast satellite		
BS-R	broadcast satellite-right hand circular polarization	BS-L	broadcast satellite – left hand circular polarization
CATV	community antenna television (network)	SCR	signal to crosstalk ratio
COFDM	coded orthogonal frequency division multiplex	CS	communication satellite
CS-L	communication satellite – left hand circular polarization	CS-R	communication satellite – right hand circular polarization
CSO	composite second order	CTB	composite triple beat
D/U	desired to undesired signal ratio	EDFA	erbium-doped fibre amplifier
E/O	optical transmitter (electrical to optical transducer)	DVB-C	digital video broadcasting baseline system for digital cable television (ETSI EN 300 429)
DVB-C2	digital video broadcasting baseline system for digital cable television second generation (ETSI EN 302 769)	DVB-S	digital video broadcasting baseline system for digital satellite television (ETSI EN 300 421)
DVB-S2	digital video broadcasting baseline system for digital satellite television second generation (ETSI EN 302 307)	DVB-T	digital video broadcasting baseline system for digital terrestrial television (ETSI EN 300 744)
DVB-T2	digital video broadcasting baseline system for digital terrestrial television second generation (ETSI EN 302 755)	FTTH	fibre to the home
GEAPON	gigabit ethernet passive optical network	GPON	gigabit passive optical network
H/E	headend	HFC	hybrid fibre coaxial
HDTV	high definition television	ISDB-C2	integrated services digital broadcasting – cable second generation
ISDB-C	integrated services digital broadcasting – cable	ISDB-S	integrated services digital broadcasting – satellite
ISDB-T	integrated services digital broadcasting – terrestrial	ISDB-S3	integrated services digital broadcasting – satellite third generation
ITU-T	International Telecommunication Union – Telecommunication sector	LD	laser diode
MDU	multiple dwelling unit	NF	noise figure
O/E	optical receiver (optical to electrical transducer)	OFDM	orthogonal frequency division multiplex
OLT	optical line terminal	OMI	optical modulation index
ONU	optical network unit	PD	photo diode

QAM	quadrature amplitude modulation	QPSK	quaternary phase shift keying
RIN	relative intensity noise	RBW	resolution bandwidth
RF	radio frequency	SBS	stimulated Brillouin scattering
SCM	single carrier modulation	SDTV	standard definition television
SDU	single dwelling unit	SMF	single mode fibre
S/N	signal-to-noise ratio	SPM	self-phase modulation
SRS	stimulated Raman scattering	TC8PSK	trellis coded 8-phase sift keying
TV	television	UHD	ultra high definition
VBW	video bandwidth	V-ONU	video optical network unit
WDM	wavelength division multiplexing	XPM	cross-phase modulation
xPSK	x-phase shift keying "x" = 4, 8, TC8, etc.		

4 Overview

This document describes the pass-through method of satellite broadcast signals over the FTTH system which uses AM-FDM (SCM) transmission. For an FTTH system below 1 GHz refer to IEC 60728-13 or IEC 60728-113. This document contains descriptions of the measurement methods and specifications for optical wavelength division multiplex and for xPSK(QPSK, 8PSK, TC8PSK) and 16APSK modulation systems. It specifies the downstream video signal transmission and thus the two-way optical transmission system is out of the scope of this document. This document applies to FTTH systems of broadband broadcast signal transmission which conveys satellite broadcast signals using one or multiple optical wavelengths. It is provided for cable/satellite operators to extend their broadband services in order to avoid interference between optical wavelengths based on the technologies that are planned to be described in IEC 60728-113.

This document also includes technical descriptions for new transmission signals, such as 4K video contents, which require more bandwidth expansion than the system described in the previous edition of this document. Furthermore, it is assumed that the 4K original video contents and other new services expected in the future (such as 8K, 3D-hologram, and/or video on demand (VOD)) will be transmitted on cable platforms via satellite link.

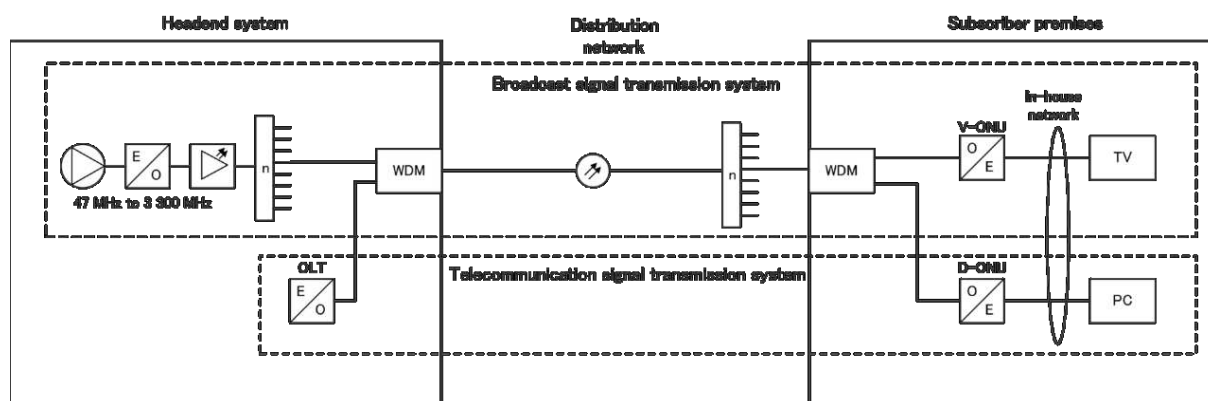
Informative Annex A describes the system composition and model system based on this document. Annex B describes basic concepts for optical wavelength division multiplexing and adds notes with respect to the system configuration. Annex C gives the minimum wavelength separation, and Annex D explains the relationship between S/N degradation and rain attenuation.

5 Optical system reference model

This clause specifies a bandwidth expansion of systems described in IEC 60728-13 and IEC 60728-113. The expansion contains a conversion method which transfers broadcast signals to optical signals using a single optical transmitter with expanded bandwidth in a one-wavelength system or by expanding the bandwidth by adding a second optical transmitter (i.e. two-wavelength system). Migrating from the original system, the following options exist:

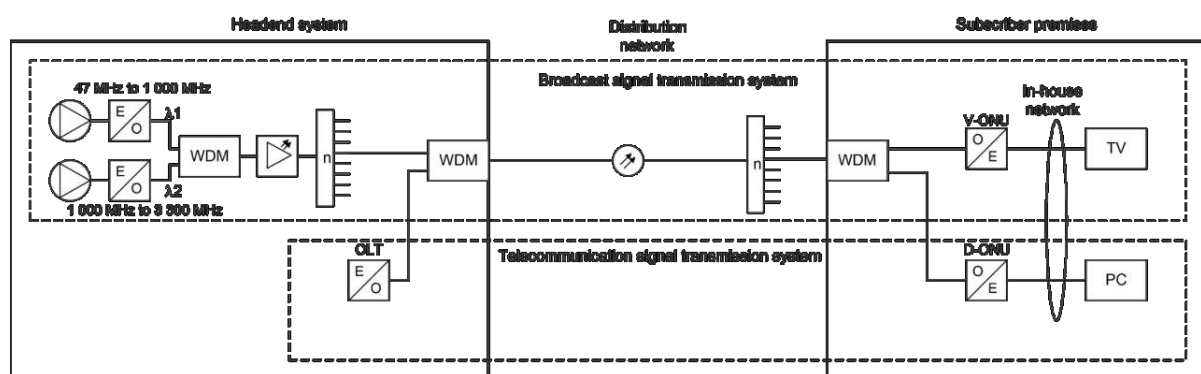
- An FTTH Cable TV system using one wavelength is shown in Figure 1.

- An FTTH Cable TV system using two wavelengths is shown in Figure 2.
- In the case of separate optical fibre transmission in broadcast and communication signals, the WDM filter in Figure 1 or Figure 2 should be removed and the system should use separated fibres for each signal.
- The two-wavelength system (Figure 2) should contain two optical transmitters and receive the two wavelength (λ_1 , λ_2) signals with a single receiver. The WDM filter at the receiving side for the separation of λ_1 and λ_2 signals is not necessary. However, the WDM filter for the separation of broadcast and communication signals is necessary at the receiving side.



IEC

Figure 1 – FTTH Cable TV system using one-wavelength

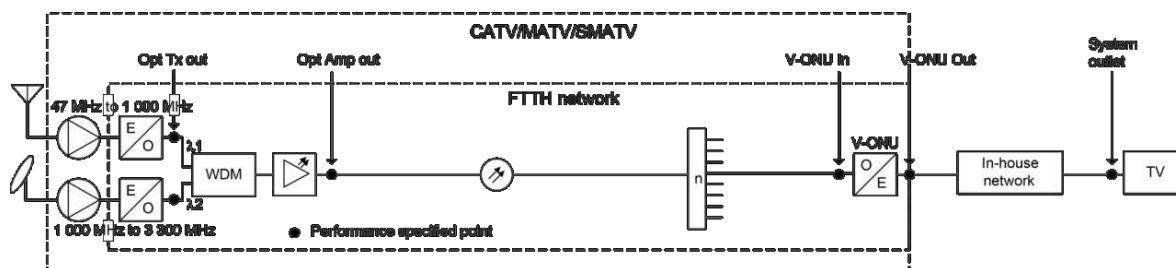


IEC

Figure 2 – FTTH Cable TV system using two wavelengths

The FTTH system below 1 000 MHz is specified in IEC 60728-13 or IEC 60728-113.

Figure 3 shows the performance specified points of the optical system.



IEC

Figure 3 – Performance specified points of the optical system

6 Preparation of measurement

6.1 Environmental conditions

6.1.1 Standard measurement conditions

Unless otherwise specified, all the measurements shall be carried out under the following standard measurement conditions.

a) Temperature and humidity

The measurement equipment shall be operated at an ambient temperature and relative humidity in the range of 15 °C to 35 °C and 25 % to 75 %, respectively, see (IEC 60068-1:2013, 4.3). Nevertheless, the specification of the measurement equipment has to be taken into account.

b) Setting up the measuring setup and system under test

The system under test shall be in the normal operating conditions, and all the pieces of equipment in the system shall be mounted and tuned according to the designed level diagram prior to the measurement.

c) AGC operation

Unless otherwise specified, all the pieces of equipment in the system shall be operated in the AGC mode if available.

d) Impedance matching between pieces of equipment

Attention shall be paid on the impedance matching between pieces of equipment and the test setup, and sufficient care shall be taken to avoid any measurement error by introducing components such as attenuators.

6.1.2 Standard operating condition

The standard operating condition refers to the condition in which the cable TV system under test is fully functional at a given facility. All the input and output of individual pieces of equipment shall be tuned according to the designed level diagram before any measurement is carried out.

6.1.3 Standard signal and measuring equipment

For measurement purposes, the standard signals used in the measuring instruments as well as in the system under test shall be set according to the specified standard signal format of the individual system. The measuring instruments to be used are described in Table 2 (passive pieces of equipment are excluded).

Table 2 – Measuring instruments

Name of instrument	Usage
Optical power meter	Instrument to measure the power of the optical signal.
Spectrum analyser	Instrument used for quantitative measurement of high frequency signals.
Optical spectrum analyser or wavelength meter	Instrument used for optical wavelength measurement
Signal generator	Instrument used to generate high frequency signals (sine-waves).
Network analyser ^a	Instrument used to measure the high frequency performance of equipment.
NF meter ^a	Instrument used to measure the noise figure (NF).
Current meter (Ammeter) ^a	Instrument used to measure electrical current.
V-ONU	Optical receiver unit used to convert an optical video signal to an electrical signal.
WDM Filter	Instrument used to separate the wavelength for measuring optical power and wavelength.
^a If the <i>RIN</i> calculation parameters of the ONU, the responsivity (<i>R</i>), the dark current (<i>I_{do}</i>) and the equivalent noise current density (<i>I_{eq}</i>) are known beforehand, these instruments are not necessary.	

6.2 Accuracy of measuring equipment

All the devices and instruments used for the measurement shall be calibrated. The standard sources used for calibration shall be calibrated within six months before the day of measurement.

6.3 Source power

The supply voltage and frequency for the measuring instruments and the equipment of the system under test shall be obtained from the corresponding instrument/equipment specifications.

7 Methods of measurement

7.1 Measuring points

The measuring points of the FTTH system for satellite broadcast transmission are illustrated in Figure 4. The methods of measurement described in Clause 7 are applied to the WDM system model and are basically similar to the system using a single wavelength.

The measurements carried out at the tap-off output can be used to predict the system performance at the output of V-ONU. Accordingly, it becomes easier for the system operators to monitor and control the performance of subscriber terminals.

It is required to measure the optical power at points (1) to (5), and the electrical signal level at points (6) and (7) of Figure 4 to ensure the total system performance. Points (5), (6) and (7) shall be measured to guarantee the system performance at the end point of the optical section and at the interface point to the customer premises. The relative intensity noise (*RIN*) shall be measured at points (1) through (5) and *S/N* (electrical signal) at points (6) and (7). An estimated signal-to-noise ratio at the output of the V-ONU is calculated from the measured *RIN* value of the optical input signal to the V-ONU at point (5).

7.3 Optical power

The measurement of optical power at single wavelengths shall be carried out according to 4.2 of IEC 60728-6:2011. For measuring the total average optical power of multiple wavelengths emanating from the end of a test fibre, the method described in IEC 60728-13 shall be used.

In general, there is no wavelength selectivity in the optical power meter that is calculated and is displayed as total optical power. Therefore, it is necessary to separate wavelengths by the WDM coupler or WDM filter. In that case, it is necessary to compensate for the loss of the WDM filter used.

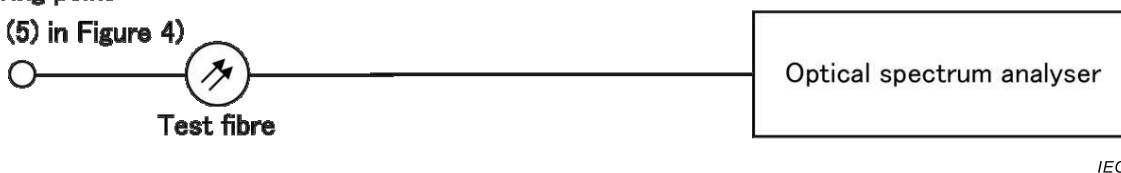
7.4 Optical wavelength

The optical wavelength, in the FTTH system for satellite broadcast signal transmission, shall be measured following the descriptions given below.

If a single V-ONU is used to receive multiple wavelengths simultaneously without any WDM filter, a test WDM filter shall be used to measure the individual optical wavelength at the input of the V-ONU. The measurement setup is shown in Figure 5a) and Figure 5b).

Measuring point

((1) to (5) in Figure 4)

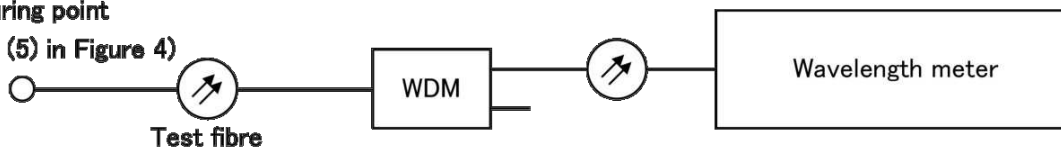


IEC

a) Measurement of optical wavelength without WDM coupler

Measuring point

((1) to (5) in Figure 4)



IEC

b) Measurement of optical wavelength using WDM coupler

Figure 5 – Measurement of optical wavelength

For measuring the central wavelength λ_0 of the spectrum of an optical signal under modulation, the method described in IEC 61280-1-3 shall be used. The central wavelength shall be expressed in nm.

7.5 Signal level and signal-to-noise ratio

7.5.1 General

The purpose of this test method is to measure the signal level of the satellite broadcast signal (16APSK, TC8PSK, 8PSK, QPSK, etc). Also, the signal-to-noise ratio is measured using the measured noise level within the transmission bandwidth of the broadcast signal. This test method performs the measurement in the electrical domain.

7.5.2 Measurement setup

Setup for the measurement of the signal level and signal-to-noise ratio is shown in Figure 6.

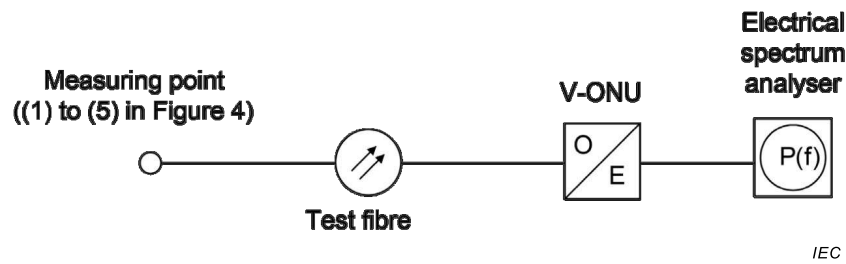


Figure 6 – Measurement of signal level and signal-to-noise ratio

The test setup shown in Figure 6 is for simplified measurement of signal-to-noise ratio in the optical portion of the network. The correct measurement should be carried out in the electrical part of the network using an electrical spectrum analyser at points (6), (7) in Table 3.

7.5.3 Measurement conditions

The following measurement conditions apply.

- The spectrum analyser used for the measurement has to be calibrated before the measurement. The supply voltage of all the pieces of equipment used for the measurements shall be switched on at least 30 min before the start of the measurement.
- If the measuring instrument has any calibration function, it shall be executed prior to the measurement.
- Suitable coaxial cables and connectors shall be used to maintain proper impedance matching within the measurement system.

7.5.4 Measurement method for xPSK signals

For the measurement of the average level of signal and signal-to-noise ratio for digitally modulated signals, the methods described in IEC 60728-1 shall be followed.

7.5.5 Presentation of the results

The signal level shall be expressed in dB(mW) or in dB(μ V) and the signal-to-noise ratio shall be expressed in dB.

7.6 *RIN* and signal-to-noise ratio

7.6.1 General

The purpose of this measurement method is to predict the signal-to-noise ratio at the output of the V-ONU from the measured relative intensity noise (*RIN*) of each used wavelength of the total optical input signal to the V-ONU.

RIN is the noise caused by fluctuations in optical output power with respect to time and is expressed as the ratio of average optical power to the average noise power measured in 1 Hz bandwidth. It is difficult to measure the *RIN* directly in the optical domain and the measurement shall be carried out after converting the optical signal to an electrical signal. However, an accurate measurement of *RIN* is not possible if the optical input to the V-ONU is small as in most of the practical systems. *RIN* may also be calculated from the measured performance of individual components constituting the system.

Considering such characteristics of *RIN*, in order to calculate the signal-to-noise ratio at the V-ONU output, it is necessary to measure the *RIN* for each wavelength at points (1) to (3), as shown in Figure 7, where the optical output power is sufficiently high to allow *RIN* measurements to be accurate.

NOTE *RIN* measurements will not be accurate when the optical power is lower than -3 dB(mW).

7.6.2 Measuring points and measurement setup

7.6.2.1 Measuring points

The measuring points for the *RIN* measurement in a typical FTTH system for satellite broadcast signal transmission are illustrated in Figure 7.

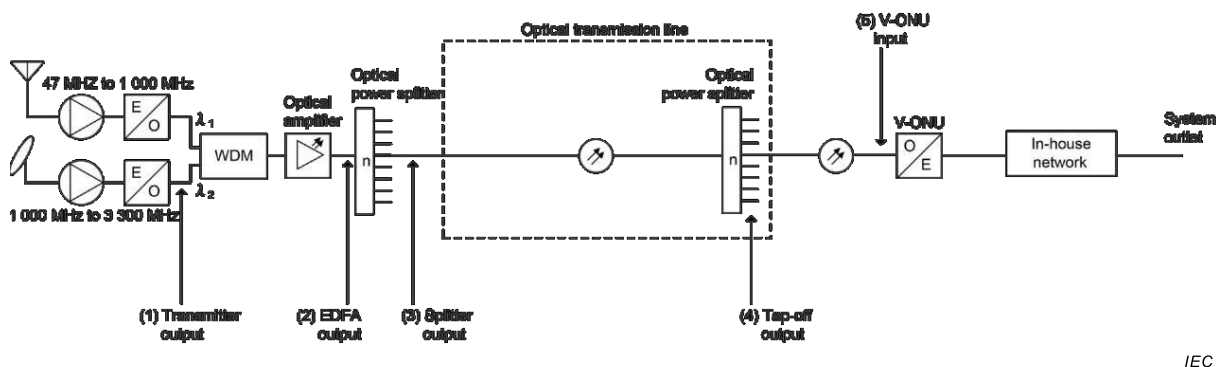


Figure 7 – Measuring points in a typical FTTH system

- If an optical amplifier is not employed in the system, the *RIN* for each wavelength shall be measured at point (1).
- Usually, optical amplifiers are installed indoors for maintenance purposes, if an outdoor type optical amplifier is employed and measurement can be carried out outdoors, the optical amplifier output shall be considered as a measuring point.
- If the optical power at points (4) and (5) are sufficiently high, these points shall also be used for measuring the *RIN* value.

7.6.2.2 Measurement setup

Figure 8 shows the *RIN* measurement setup.

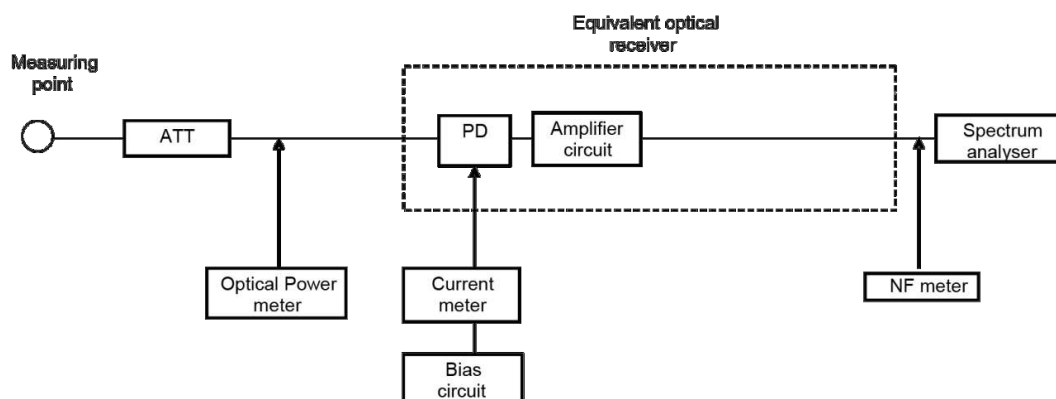


Figure 8 – *RIN* measurement setup

7.6.3 Measurement conditions

The measurement conditions of *RIN* measurement are the same as those given in 6.4.3 of IEC 60728-13:2010.

7.6.4 System *RIN* measurement method

7.6.4.1 General

This test method shall be used to predict the signal-to-noise ratio at the output of a V-ONU from the *RIN* measurement using the setup shown in Figure 8. If multiple wavelengths are used in the system, *RIN* shall be measured at all the individual wavelengths. In order to measure *RIN* at individual wavelengths, either an optical wavelength filter shall be inserted at the measuring point or the transmitters of other wavelengths shall be turned off. Ensure that only the wavelength of interest is fed to the measurement setup.

This sub-clause contains several steps as described below. If the parameters, R , I_{d0} , I_{eq} and G are unknown, refer to Annex D of IEC 60728-13:2010 for the measurement method of these parameters. *RIN* can be calculated using these parameters.

7.6.4.2 STEP A: Input power of optical receiver and system noise (noise current density)

For step A proceed as follows.

- Measure the input power of the optical receiver (P_r) using a power meter.
- Connect the spectrum analyser to the output of the optical receiver and select the measurement mode to measure the noise power density. Measure the noise power density per unit frequency, N_p expressed in dB (mW)/Hz. The total noise current per Hz, I_{bn} of the optical receiver can be calculated using Equation (1) with the RBW of the spectrum analyser set to 100 kHz.

$$I_{bn} = \sqrt{\frac{N_p}{Z_0} \times 10^{-3}} \quad [A/\sqrt{Hz}] \quad (1)$$

where

Z_0 is the impedance of the measurement setup,

N_p is the noise power density, expressed in dB(mW)/Hz.

The following correction shall be applied if the noise level (N_L) is measured with the spectrum analyser:

$$N_p = N_L + 10 \lg\left(\frac{B_N}{B}\right) + K_1 + K_2 \quad (2)$$

where

B_N is the measurement bandwidth of noise power (N_p) in Hz,

B is the noise bandwidth, $RBW \times 1,2$ (noise bandwidth correction factor), 120 000 Hz,

K_1 is the correction factor for the conversion to effective voltage level,
 $20 \lg(2/\sqrt{\pi}) = 1,05$ dB,

K_2 is the correction factor for the logarithmic amplifier of the spectrum analyser, typically 1,45 dB.

The measured noise level (N_p) includes that of the measuring equipment (spectrum analyser) which should be at least 20 dB lower than the noise level displayed outside the channel band in order not to affect the results. Otherwise, the contribution of noise (due to the system or the equipment under test and to the measuring equipment) shall be taken into account in the measurement of noise level (see Annex F of IEC 60728-1:2014).

7.6.4.3 STEP B: *RIN* calculation

For step B proceed as follows.

- From the above measurement results, *RIN* can be calculated from the following relation:

$$RIN_n = 10 \lg \left(\frac{\frac{I_{bn}^2}{G}}{(R \cdot P_{rn})^2} - \frac{2e}{(R \cdot P_{rn})^2} (I_{d0} + R \cdot P_{rn}) - \frac{I_{eq}^2}{(R \cdot P_{rn})^2} \right) \left[\text{dB}(\text{Hz}^{-1}) \right] \quad (3)$$

where

- R* is the responsivity of the photodiode (A/W),
- I_{d0}* is the dark current of the photodiode (A),
- I_{eq}* is the preamplifier equivalent input noise current density (A/√Hz),
- I_{bn}* is the total noise current within 1 Hz bandwidth at the optical receiver output (A/√Hz),
- G* is the amplifier gain of the optical receiver (Including gain of matching circuit),
- P_{rn}* is the *n*-th input power to the optical receiver (W),
- e* is the charge of an electron $1,602 \times 10^{-19}$ (C).

7.6.5 *S/N* calculation based on *RIN* value

The signal-to-noise ratio (*S/N*)_{*sk*} value of the *k*-th RF signal in the *s*-th optical signal contained in *n* (1 to *N_T*) optical signals, at the V-ONU output can be calculated using the following relation:

$$(S/N)_{sk} = 10 \lg \left(\frac{1}{B_N} \cdot \frac{\frac{1}{2} \cdot (OMI_{sk} \cdot R \cdot P_{rs})^2}{\sum_{n=1}^{N_T} \{ RIN_n \cdot (R \cdot P_{rn})^2 \} + 2 \cdot e \cdot \left(I_{d0} + \sum_{n=1}^{N_T} R \cdot P_{rn} \right) + I_{eq}^2} \right) \left[\text{dB} \right] \quad (4)$$

The parameters for the calculation are listed in Table 4.

Table 4 – Parameters used to calculate S/N when signals of multiple wavelengths are received by a single V-ONU

Symbol	Description		Remarks
B_N	ISDB-C	Noise bandwidth 64/256-QAM: $5,3 \times 10^6$ (Hz)	Depends on the modulation format
	ISDB-C2	OFDM(J.382): $5,71 \times 10^6$ (Hz)	
	ISDB-T	OFDM: $5,6 \times 10^6$ (Hz)	
	ISDB-S	QPSK, TC8PSK: $28,86 \times 10^6$ (Hz)	
	ISDB-S3	16APSK: $33,76 \times 10^6$ (Hz)	
K_s	Total number of RF signals in s -th optical signal		
OMI_{sk}	Optical modulation index of k -th RF signal transmitted using the s -th optical signal		
P_{rs}	Optical power of the s -th optical (W)		To measure the optical power of any individual wavelength, either turn off the transmitters of all other wavelengths, or insert an optical wavelength filter at the measuring point, and ensure that only the wavelength of interest is fed to the power meter.
P_{rn}	Optical power of n -th optical signal(W)		This parameter depends on the transmission line design.
RIN_n	RIN of the n -th optical (dB(Hz ⁻¹))		This parameter depends on the optical transmitter, optical amplifiers and transmission lines. If these parameters are unknown, the following values may be used to calculate the RIN of optical signal input to the V-ONU. RIN of the optical transmitter for multi-channel transmission: -155 dB(Hz⁻¹). RIN of the optical transmitter for retransmission: -150 dB(Hz⁻¹). NF of an optical amplifier: 6,5 dB. RIN due to the optical transmission line: -161 dB(Hz⁻¹).
e	Charge of an electron (1,602 x 10 ⁻¹⁹ C)		Physical constant.
R	Responsivity of the V-ONU (A/W)		Depends on the performance of V-ONU. If these parameters are unknown, the following values may be used in the calculation of RIN .
I_{d0}	Dark current of the V-ONU (A)		
I_{eq}	V-ONU equivalent input noise current density (A/√Hz)		
N_T	Number of simultaneously transmitted optical signals		R : 0,84 (A/W) I_{d0} : 1,3 (nA) I_{eq} : 10 pA/√Hz

7.6.6 Calculation of component RIN

To calculate the RIN degradation due to optical amplifiers and optical transmission lines, the methods described in 6.4.6 of IEC 60728-13:2010 shall be used.

7.7 Optical modulation index

The optical modulation index (*OMI*) of broadcast satellite signals shall be measured according to the method described in 4.8 of IEC 60728-6:2011. In this document it is required that any power AGC function, if available in the transmitter, shall be turned off during the measurements.

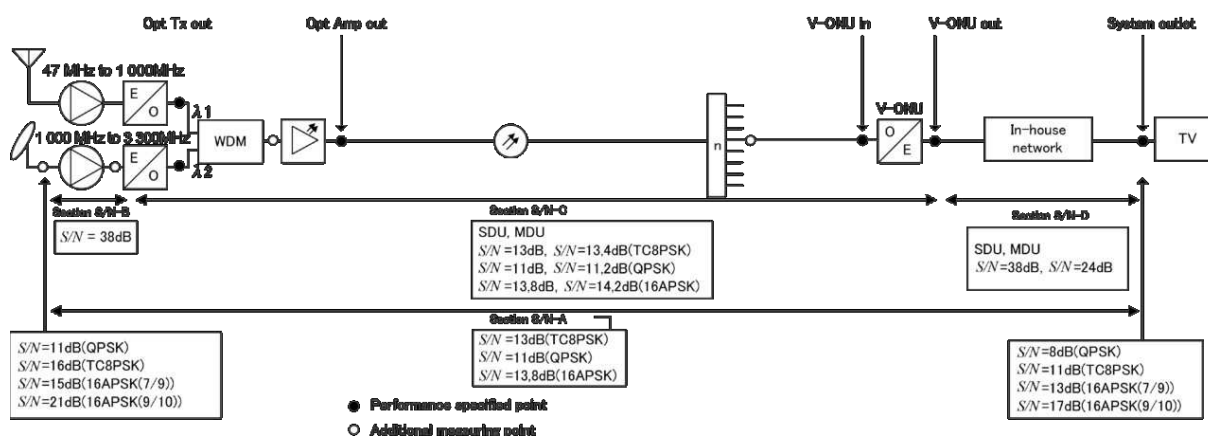
7.8 Signal-to-crosstalk ratio (*SCR*)

This method of measurement is applicable when other services (i.e. digital communication signals like GPON, GEAPON or Ethernet-Point-to-Point) besides CATV broadcast transmission (i.e. 16APSK, 64/256QAM, OFDM, 8PSK and QPSK) are transmitted in the optical network. Other services may produce crosstalk effects in optical fibres and in optical receiver devices with high linearity. The signal-to-crosstalk ratio (*SCR*) of satellite broadcast signals shall be measured according to the method of measurement of signal-to-crosstalk ratio which is planned to be described in 6.6 of IEC 60728-113:—.

8 Specification of optical system for broadcast signal transmission

8.1 Digital broadcast system over optical network

OFDM and QAM (47 MHz to 1 000 MHz) systems are described in IEC 60728-113. This document describes broadcast satellite (BS) and communication satellite (CS) signals (1 000 MHz to 3 300 MHz) modulated by 16APSK and xPSK. The following *S/N* allocation is applied only to these signals (see Figure 9).



IEC

Figure 9 – Performance allocation and measuring points

8.2 International TV systems

The minimum RF signal-to-noise requirement in operation is shown in Table 5.

Table 5 – Minimum RF signal-to-noise ratio requirements in operation

System	Modulation	Code rate	Minimum RF signal-to-noise ratio at headend input	Minimum RF signal-to-noise ratio at system outlet
			S/N dB	S/N dB
ISDB-S	TC8PSK	2/3	16 ^a	11
	QPSK	3/4	11 ^a	8
ISDB-S3	16APSK	7/9	21 ^a	13 ^a
DVB-S	QPSK	3/4	See Table 17 of IEC 60728-101:2016	See Table 9 of IEC 60728-101:2016
DVB-S2	16APSK	—	See Table 17 of IEC 60728-101:2016	See Table 9 of IEC 60728-101:2016
	32APSK	—	See Table 17 of IEC 60728-101:2016	See Table 9 of IEC 60728-101:2016

^a It shows the subtraction of “corresponding S/N degradation due to rain attenuation of 99 % in time at the worst month” from “ S/N of the headend input signal”. Refer to Annex D for details of the relation between S/N degradation and rain attenuation.

8.3 Relationship between RIN and S/N

The CATV broadcast service within the scope of this document can be classified into three types:

- metropolitan type CATV;
- municipal type CATV;
- poor signal reception type CATV.

The broadcast channels can be transmitted using either a single or dual (WDM) wavelengths.

Table 6 shows the types of broadcast services with the typical number of signals. Annex A should be referred to for the combination of digital signals and satellite signals in actual system design.

The multi-channel service system, for longer transmission distance and larger number of subscribers, uses a transmitter with the external intensity modulation method for the transmission of digital signals and satellite signals. In multi-channel service with CS supplementary service, the BS/CS-IF channels are transmitted using the direct intensity modulation method.

The re-transmission service system is a small-sized receiving facility with poor reception of broadcast TV programs, nine digital signals and BS service are assumed for the re-transmission service. Most of the re-transmission service systems don't require an EDFA, or require one EDFA only, and the optical system uses the direct intensity modulation method in general.

Table 6 – Types of broadcast services

Type of service	Number of wavelengths		Signal less than 1 GHz				Higher than 1 GHz (satellite) ISDB-S/S3 DVB-S/S2
			Number of signals	System	Modulation	Sub-carrier	
Metropolitan type CATV	1		80 signals	DVB-T (ISDB-T)	COFDM (OFDM)	64QAM	24 signals (xPSK, 16APSK)
				DVB-T2	COFDM	~ 256QAM	
				DVB-C (ISDB-C)	64QAM 256QAM	— —	
				DVB-C2 (J.382)	COFDM (OFDM)	~ 4096QAM	
	2	λ_1	80 signals	DVB-T (ISDB-T)	COFDM (OFDM)	64QAM	—
				DVB-T2	COFDM	~ 256QAM	
				DVB-C (ISDB-C)	64QAM 256QAM	— —	
				DVB-C2 (J.382)	COFDM (OFDM)	~ 4096QAM	
		λ_2	—	—	—	—	24 signals (xPSK, 16APSK)
Municipal type CATV	1		11 signals	DVB-T (ISDB-T)	COFDM (OFDM)	64QAM	48 signals (xPSK, 16APSK)
				DVB-T2	COFDM	~ 256QAM	
	2	λ_1	11 signals	DVB-T (ISDB-T)	COFDM (OFDM)	64QAM	—
				DVB-T2	COFDM	~ 256QAM	
		λ_2	—	—	—	—	48 signals (xPSK, 16APSK)
Poor signal reception type CATV	1		9 signals	DVB-T (ISDB-T)	COFDM (OFDM)	64QAM	24 signals (xPSK, 16APSK)
				DVB-T2	COFDM	~ 256QAM	
	2	λ_1	9 signals	DVB-T (ISDB-T)	COFDM (OFDM)	64QAM	—
				DVB-T2	COFDM	~ 256QAM	
		λ_2	—	—	—	—	24 signals (xPSK, 16APSK)

The performance of such a transmission system can be defined by the value of relative intensity noise (RIN) for the optical signal, and the S/N ratio at V-ONU output for the electrical signal. The term V-ONU is used as the synonym of optical receiver (O/E) device in this document, and in the case of services employing dual-wavelength, the multiplexed signal is assumed to be received by a single V-ONU. Details on actual parameters are described in Annex A. RIN values required

for the three service types are shown in Table 7. The intensity modulation method is applied to the optical system in all service types.

The following tables are applied only to the broadcasting satellite signal band using digital modulation (16APSK, TC8PSK and QPSK). The *RIN* value is calculated based on the necessary *S/N* ratio for the transmission of enhanced broadcasting signals (broadcast satellite digital signals). The overall system *S/N* shall be allocated based on the digital signals in which the more stringent condition is required.

Table 7 – Type of service and minimum operational *RIN* values for satellite services

Type of service	Number of wavelengths	OMI %	V-ONU minimum input level dB(mW)	System <i>RIN</i> minimum value dB(Hz ⁻¹)	Corresponding <i>S/N</i> value ^a dB
Metropolitan type CATV (type a)	1	1,5	−8	−128 −129	13 (SDU) 14 (MDU)
	2	—	−8 (λ_1) ^b	−128 ^c	13 (SDU)
		6,1	−14 (λ_2) ^b	−129 ^c	14 (MDU)
Municipal type CATV (type b)	1	3,0	−8	−127 −128	18 (SDU) 19 (MDU)
	2	—	−8 (λ_1) ^b	−138 ^c	18 (SDU)
		4,3	−14 (λ_2) ^b	−139 ^c	19 (MDU)
Poor signal reception type CATV (type c)	1	4,7	−12	−118 −119	13 (SDU) 14 (MDU)
	2	—	−12 (λ_1) ^b	−118 ^c	13 (SDU)
		6,1	−12 (λ_2) ^b	−119 ^c	14 (MDU)
^a The <i>S/N</i> value is calculated for TC8PSK system (for type of service a, c) or 16APSK(Code Rate 7/9) system (for type of service b).					
^b λ_1 : wavelength of the 47 MHz to1 000 MHz transmission band, λ_2 : wavelength of the 1 000 MHz to 3 300 MHz transmission band (for type of service a, c) or 1 000 MHz to 3 300 MHz (for type of service b).					
^c Total <i>RIN</i> value for a two-wavelength system.					

Multiple optical reflections over the transmission line may degrade the *RIN* values. In order to minimize this degradation, the use of grade 2 connectors according to IEC 61755-1 or APC optical connectors is recommended.

8.4 Optical wavelength

Table 8 shows the performance of optical wavelength and power for the FTTH system defined by this document.

Table 8 – performance of optical wavelength and power

Optical wavelength		Refer to IEC 60728-13.
Optical power	Optical transmission power	When the FTTH system operates with two wavelengths, the optical power of the first wavelength refers to IEC 60728-13 and the second wavelength is -6 dB of the first wavelength.
	V-ONU input power	
Interval of wavelengths (two wavelength system)		The wavelengths shall be selected in accordance with ITU-T G.694.1 for the DWDM case and ITU-T G.694.2 for the CWDM case (see Annex B). If other wavelengths are chosen, at least 0,3 nm wavelength separation shall be kept in any case to avoid interference to broadcast signals.

Avoiding the influence to the first wavelength from the second one, the level of the second wavelength shall be sufficiently less than the first one. This means -6 dB level difference when the *OMI* of the second signal is quadrupled. Refer to B.3.2.

8.5 Frequency of source signal

The frequency range of source signals considered here is 47 MHz to 3 300 MHz. However, regional frequency plans can be used for the operating frequency range of the optical system.

8.6 Optical system specification for satellite signal transmission

The major parameters of the optical system are shown in Table 9. Refer to IEC 60728-113 for OFDM, and QAM signals.

Table 9 – Optical system specification

Parameter		QPSK (ISDB-S)	TC8PSK (ISDB-S)	16APSK (ISDB-S3)
Optical wavelength		1 530 nm to 1 625 nm (1 555 nm $\pm$ 5 nm is strongly recommended.)		
Frequency of source signal		1 000 MHz to 3 300 MHz		
Fluctuation of signal-wave level		—		
Noise bandwidth of electronic signal		28,86 MHz	28,86 MHz	33,7561 MHz
Min. optical section <i>S/N</i> (Between headend output and system outlet)		16 dB	16 dB	Code rate 7/9 15,3 dB
Min. <i>S/N</i>	At headend Input	11 dB ^a	16 dB ^a	Code rate 7/9 21 dB ^a
	At system outlet ^b	8 dB ^b	11 dB ^b	Code rate 7/9 13 dB ^c
^a It shows the subtraction of "corresponding <i>S/N</i> degradation due to rain attenuation of 99 % in time at the worst month" from <i>S/N</i> of the headend input signal. Refer to Annex D for details of the relation between <i>S/N</i> degradation and rain attenuation. ^b It corresponds to $BER = 1 \times 10^{-8}$ ^c It corresponds to $BER = 1 \times 10^{-11}$				

8.7 S/N ratio specification for in-house and in-building wirings

The S/N ratio can be specified outside of the system outlet if the performance of the in-house/in-building wiring section is maintained properly. Based on current installation methods, the S/N ratio allocation for the in-house/in-building wiring section is specified in Table 10. The S/N ratio allocation is different for single dwelling units (SDU) and multiple dwelling unit (MDU) as shown in Figure 10 and Figure 11, considering difference of network composition.

Table 10 – Section of S/N ratio specification for in-house/in-building wiring

Category of house		Minimum S/N ratio for in-house/in-building wiring dB
Single dwelling unit (SDU)	-	38
Multiple dwelling unit (MDU)	O/E conversion at MDU entrance, coaxial cable distribution to TV set	24
	No O/E conversion at MDU entrance, optical cable distribution to TV set	24

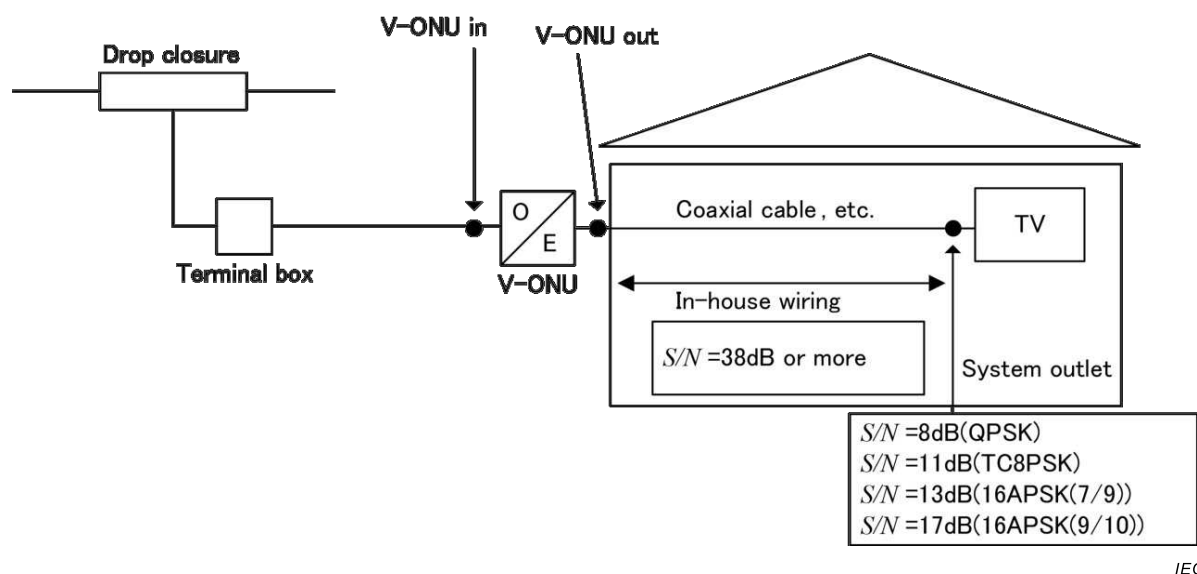
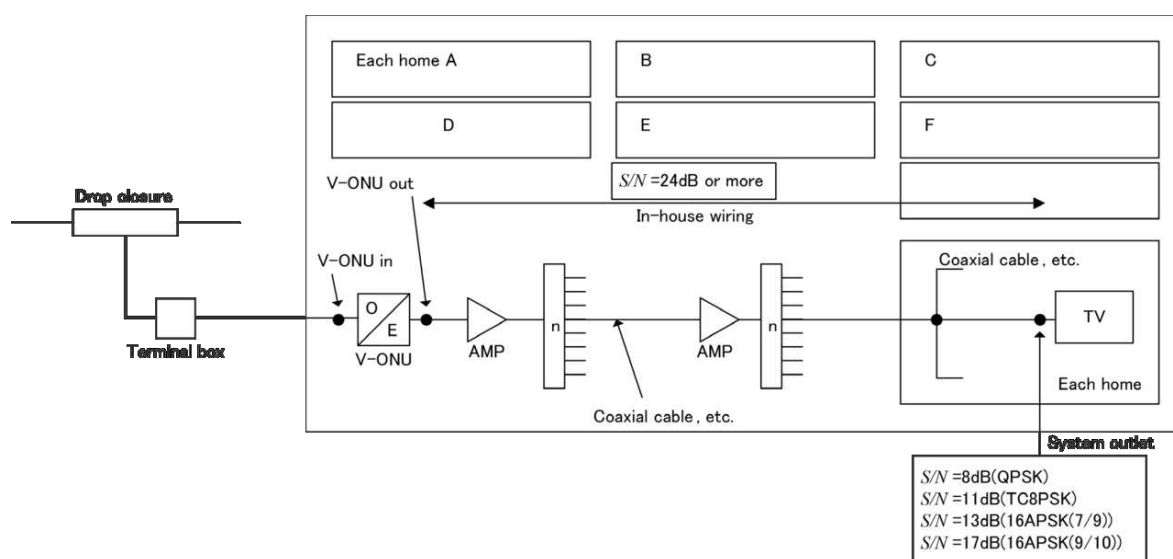


Figure 10 – Section of S/N ratio specification (38 dB) for in-house wiring



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Figure 11 – Section of S/N ratio specification (24 dB) for in-building wiring (in case of coaxial cable distribution after V-ONU)

8.8 Crosstalk due to optical fibre non-linearity

Refer to IEC 60728-113.

8.9 Single frequency interference level due to fibre non-linearity

The single frequency interference level caused by fibre non-linearity shall meet the following values shown in Table 11. All the parameters of optical broadcast transmission systems shall be set appropriately to satisfy the interference level. The measuring point shall be point (6) shown in Figure 4.

Table 11 – Interference level due to fibre non-linearity (single frequency interference)

System	Broadcast system	Code rate	D/U ratio dB
ISDB-S	Broadcast satellite (TC8PSK)	—	More than 13
	Communication satellite (QPSK)	—	More than 13
ISDB-S3	Broadcast satellite (16APSK)	7/9	More than 14
	Communication satellite(16APSK)	7/9	More than 14

8.10 Environment condition

Refer to IEC 60728-13.

Annex A

(informative)

Actual service systems and design considerations

A.1 General

As described in Clause 7, the CATV broadcast service and new services like a 3D-hologram and/ or video on demand (VOD) under the scope of this document can be classified into the following three types:

- metropolitan type CATV: Multi-channel service with digital signals (below 1 GHz, 80 channels);
- municipal type CATV: Multi-channel service with a mixture of BS,CS services and digital signals (below 1 GHz, 11 channels);
- poor signal reception type CATV: BS signals and digital signals (below 1 GHz, 9 channels).

Annex A describes actual service systems and design considerations based on the specifications stated in this document.

A.2 Metropolitan type CATV

CATV operators currently provide multi-channel services for digital broadcasting. Digital multi-channel services are classified as multi-channel services in this document. Clause A.2 describes the following two reference models for the multi-channel services over an optical network using a single optical wavelength or dual optical wavelengths, respectively.

Model (A) λ_1 : 80 digital signals + 24 satellite (BS) channels.

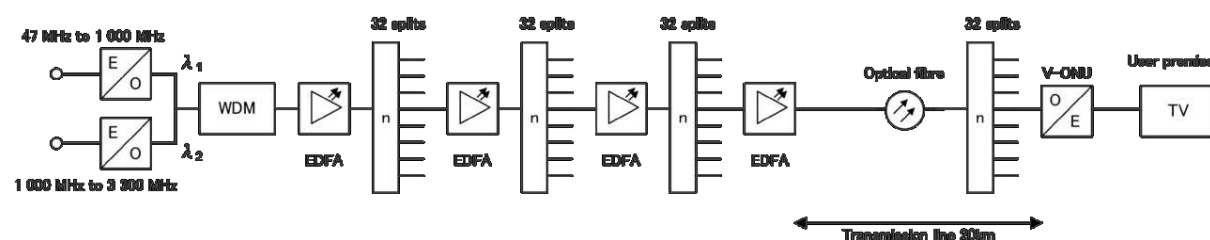
Model (B) λ_1 : 80 digital signals, λ_2 : 24 satellite (BS) channels.

Model (A) is selected as a reference model for a single optical wavelength. In this case, a directly modulated transmitter is used.

Model (B) is selected as a reference model that uses two optical wavelengths. In this case an externally modulated transmitter is used for below 1 000 MHz and a directly modulated transmitter is used for above 1 000 MHz.

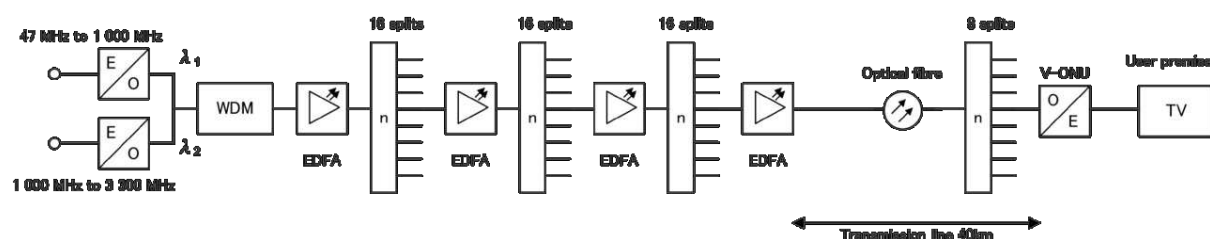
“Digital” means either DVB-C2 (J.382), 64/256QAM or COFDM (OFDM) signals, and “satellite”, 16APSK and xPSK signals.

The system size of CATV multi-channel services is mostly from 1 500 to 340 000 terminals, the transmission line length is up to 40 km, and contains four stages of EDFA in general. EDFA is the synonym of optical amplifier in this document. In the optical network system, the relationship between transmission distance and the number of branches is in inverse proportion. If the number of branches is reduced, the optical line is extendable up to 40 km, and one million or more subscriber terminals become available by stacking splitters. Figure A.1 and Figure A.2 show the examples of the multi-channel service system for one million terminals, and for 2 000 terminals, respectively.



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Figure A.1 – Example of a multi-channel service system of one million terminals



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Figure A.2 – Example of a multi-channel service system with 2 000 terminals

A.3 Municipal type CATV

Clause A.3 describes the following two reference models for the multi-channel services with CS supplementary service over an optical network using a single optical wavelength or two optical wavelengths, respectively (see Figure A.3).

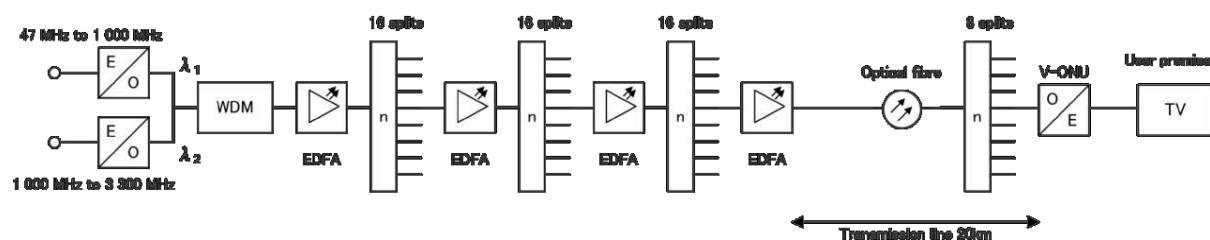
Model (C) λ_1 : 11 digital channels + 48 satellite (BS and CS) channels.

Model (D) λ_1 : 11 digital channels, λ_2 : 48 satellite (BS and CS) channels.

Model (C) is selected as a reference model for a single optical wavelength. In this case, a directly modulated transmitter is used.

Model (D) is selected as a reference model that is used for two optical wavelengths. In this case, a directly modulated transmitter is used for below 1 000 MHz and a directly modulated transmitter is used for above 1 000 MHz.

“Digital” means either 64/256QAM or OFDM signals, and “satellite”, either 16APSK or xPSK signals.



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Figure A.3 – Example of a multi-channel with CS supplementary service system for 2 000 terminals

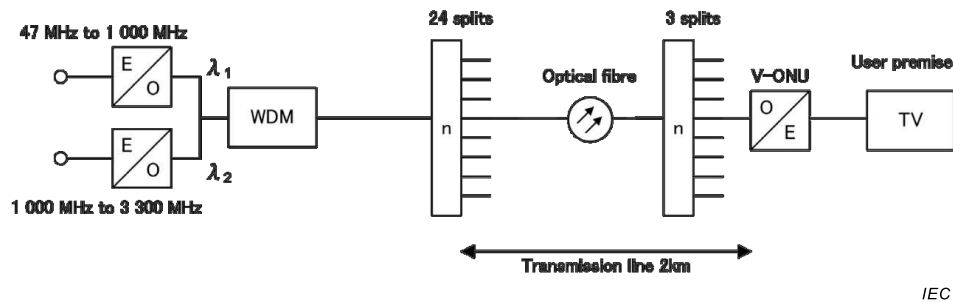
A.4 Poor signal reception type CATV

The re-transmission service system is a small-sized receiving facility providing broadcast services for poor reception areas. Nine digital channels, and 24 satellite channels are assumed for the re-transmission service. In general re-transmission service systems have about 2 km trunk line and 70 terminals. In this case, a maximum of one EDFA is enough for this network size. Most of the re-transmission service systems do not require an EDFA (see Figures A.4 and A.5).

Model (E) λ_1 : 9 digital channels + 24 satellite (BS) channels.

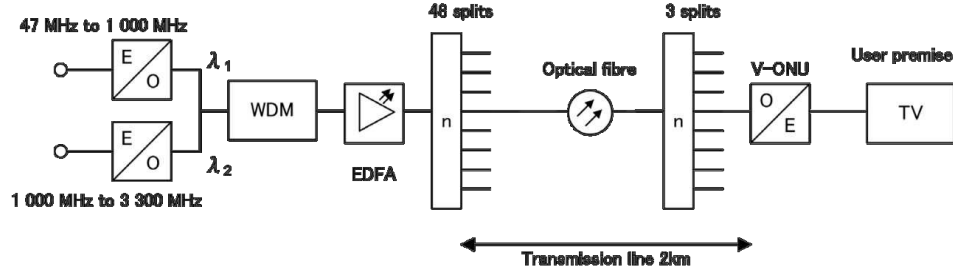
Model (F) λ_1 : 9 digital channels, λ_2 : 24 satellite (BS) channels.

Model (E) is selected as a reference model for single optical wavelength systems, and Model (F) for dual optical wavelength systems. In both cases, directly modulated transmitters are used.



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Figure A.4 – Example of a re-transmission service system with 72 terminals



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Figure A.5 – Example of a re-transmission service system with 144 terminals

A.5 System reference model

A.5.1 System parameters

Table A.1 summarizes the basic system parameters verified for the above service system and Figures A.6 to A.11 provide system performance calculations for the different models.

Table A.1 – Basic system parameters (Japan)

Ref. model No.	Type of service	Number of wave-lengths	Number of signals	OMI (%)	EDFA stage	Remark
Model A	Metropolitan type CATV	1	Digital: 80 channels	OFDM(ISDB-T): 2,5 64QAM(ISDB-C): 2,5 256QAM(ISDB-C): 5,0 Satellite (BS): 1,5	4	Direct modulation method (90 MHz to 2 681 MHz)
Model B		2	Satellite (BS): 24 channels	OFDM(ISDB-T): 2,5 64QAM(ISDB-C): 2,5 256QAM(ISDB-C): 5,0 Satellite (BS): 6,1	4	External modulation method (90 MHz to 1 000 MHz) Direct modulation method (1 000 MHz to 2 681 MHz)
Model C	Municipal type CATV	1	Digital: 11 channels	OFDM(ISDB-T): 3,2 Satellite (BS and CS): 3,0	3	Direct modulation method (90 MHz to 3 224MHz)
Model D		2	Satellite (BS and CS): 48 channels	OFDM(ISDB-T): 3,2 Satellite (BS and CS): 4,3	3	Direct modulation method (90 MHz to 1 000 MHz) Direct modulation method (1 000 MHz to 3 224 MHz)
Model E	Poor signal reception type CATV	1	Digital: 9 signals	OFDM(ISDB-T): 6,4 Satellite (BS): 4,7	0	Direct modulation method (90 MHz to 2 681 MHz)
Model F		2	Satellite (BS): 24 signals	OFDM(ISDB-T): 6,4 Satellite (BS): 6,1	0	Direct modulation method (90 MHz to 2 681 MHz)

The parameters used for the calculation of the system performance are as follows.

1) Metropolitan / Municipal type CATV

Connection loss at connector	0,5 dB/point
Fibre loss including splicing loss	0,35 dB/km
V-ONU equivalent input noise current density (I_{eq})	10 pA/ $\sqrt{\text{Hz}}$
V-ONU receiving device dark current (I_{d0})	1,3 nA
V-ONU optical-electrical conversion efficiency (R)	0,84 A/W

2) Poor signal reception type CATV

Connection loss at connector	0,5 dB/point
Fibre loss including splicing loss	0,35 dB/km
V-ONU equivalent input noise current density (I_{eq})	8,3 pA/ $\sqrt{\text{Hz}}$
V-ONU receiving device dark current (I_{d0})	1,0 nA
V-ONU optical-electrical conversion efficiency (R)	0,9 A/W

A.5.2 Operating environment

The optical transmitter and the optical amplifier are assumed to be headend equipment and installed in an office building with controlled environmental conditions. The V-ONU can be

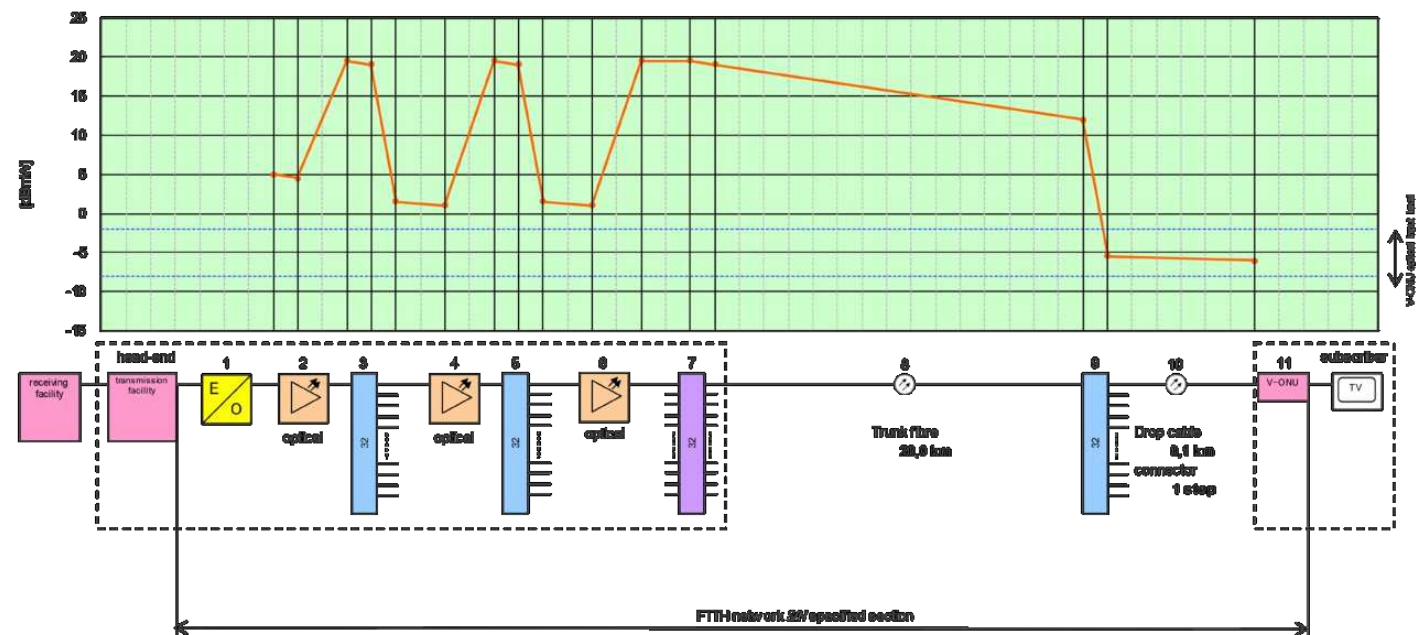
installed indoors or outdoors under eaves. Unless otherwise specified, the following ranges shall be applied.

a) Optical transmitter		
Ambient temperature	0 °C to +40 °C	
Humidity	20 % to 90 %	without dew condensation (see note)
b) Optical amplifier		
Ambient temperature	0 °C to +40 °C	
Humidity	20 % to 90 %	without dew condensation (see note)
c) V-ONU		
Ambient temperature	–20 °C to +40 °C	(outdoor installation)
Ambient temperature	0 °C to +40 °C	(indoor installation)
Humidity	20 % to 100 %	without dew condensation (see note)
Ingress protection	better than IP43	[IEC 60529]

NOTE Except direct solar radiation environment.

Model A

Metropolitan type CATV, 4 EDFA stages, OMI: OFDM/64QAM: 2,5 %, 256 QAM: 5,0 %, satellite (BS) 1,5 %
directly modulated optical transmitter: **1 wavelength** (80 digital channels, 24 satellite (BS) channels)



Item	
amplifier NF	6,2 dB
OMI	2,5 % 5,0 % 1,5 %
RF output level	27,8 dBm 43,8 dBm 50,8 dBm

equipment	1	2	3	4	5	6	7	8	9	10	11
transmitter	RF-20dBm/V	4,50	19,00	1,00	19,00	1,00	19,00	19,00	12,00	-5,50	-0,04
input level [dBm/V]	-	19,00	-17,50	19,00	-17,50	19,00	19,00	-0,50	-7,00	-17,50	-
output level [dBm/V]	5,00	19,00	1,00	19,00	1,00	19,00	19,00	12,00	-5,50	-0,04	RF-20dBm/V
output RF-40dB (1c-1)	-190,10	-102,73	-	-149,00	-	-145,00	-	-	-	-	-
channel SNR [dB] note1	-	-	-	-	-	-	-	-	-	-	30,06
channel SNR [dB] note1	-	-	-	-	-	-	-	-	-	-	48,82
channel SNR [dB] note1	-	-	-	-	-	-	-	-	-	-	28,10

note 1: FTTH network SNR at the V-ONU RF output, which is calculated from RF of the optical transmitter and the amplifier

note 2: SNR(OFDM/64QAM:SNR1,256QAM:SNR2,BS:SNR3) at V-ONU RF output, which is calculated from FTTH network SNR

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Figure A.6 – System performance calculation for model A

Model B

Metropolitan type CATV, 4 EDFA stages, OMI: OFDM/64QAM: 2,5%, 256QAM: 5,0%, satellite(BS): 6,1% externally and directly modulated optical transmitters: **2 wavelengths** (80 digital channels, 24 satellite(BS) channels)

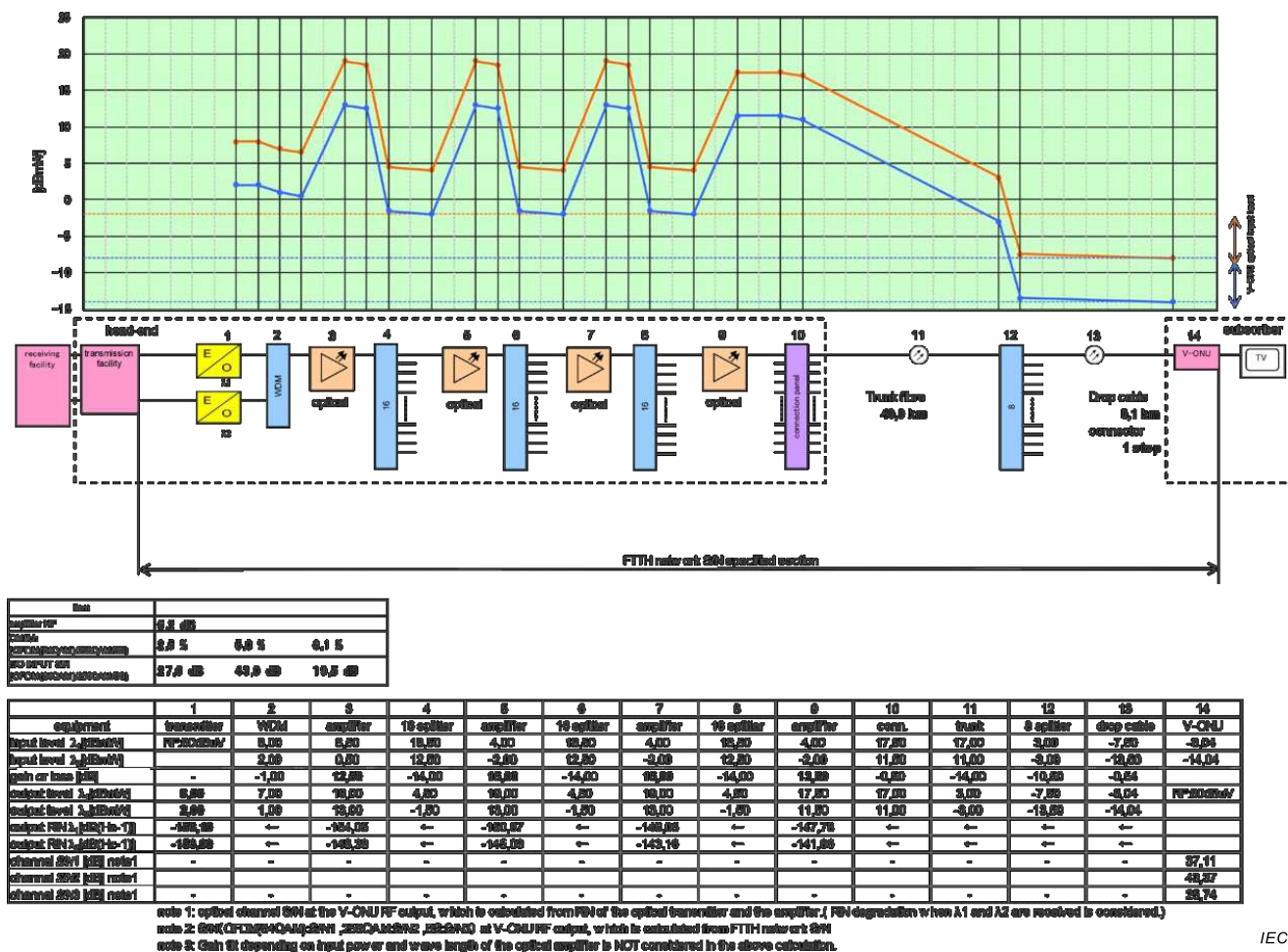
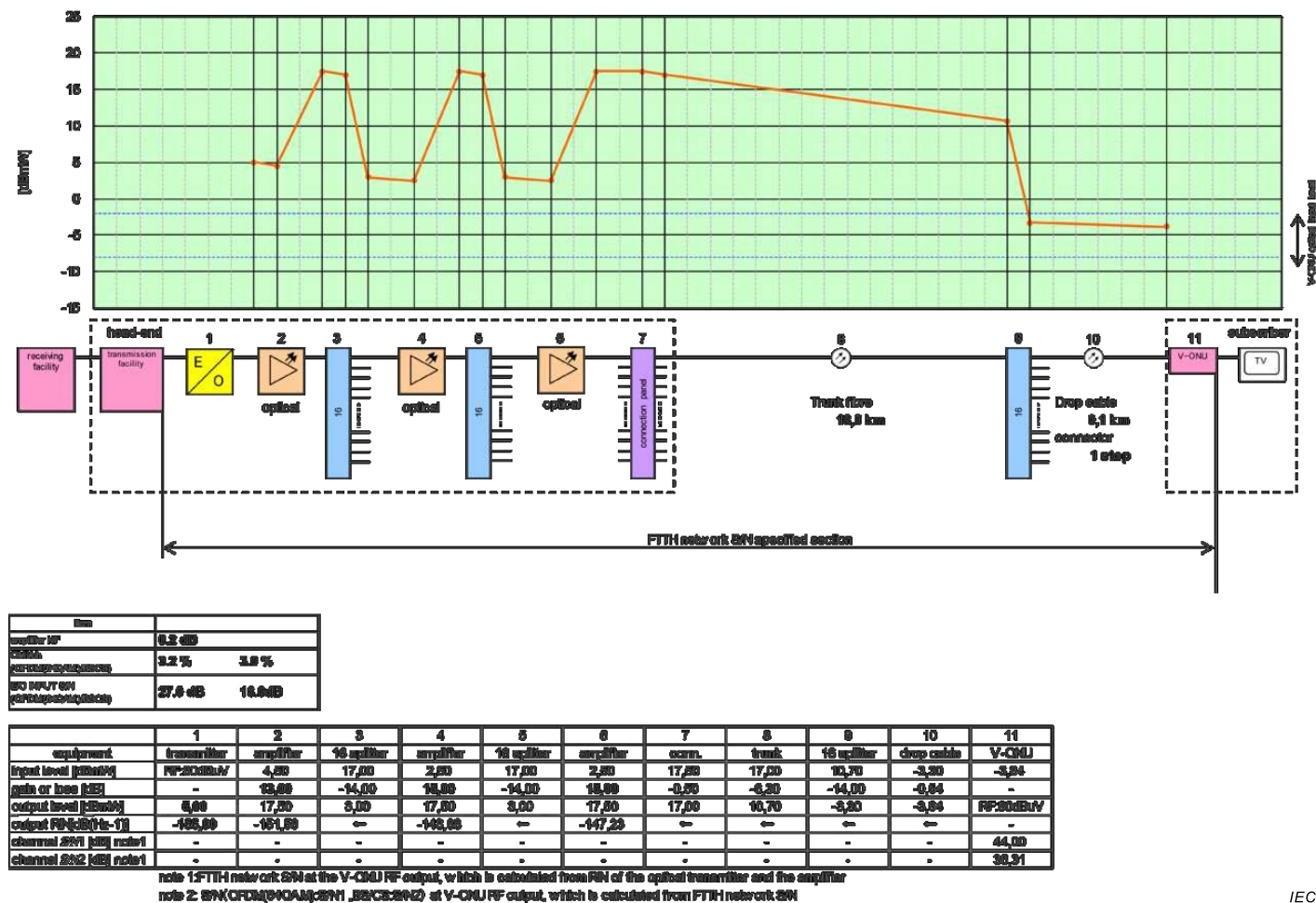


Figure A.7 – System performance calculation for model B

Model C

Municipal type CATV, 3 EDFA stages, OMI: OFDM 3,2 %, satellite (BS/CS) 3,0%
 directly modulated optical transmitter: **1 wavelength** (11 digital channels, 48 satellite (BS/CS) channels)

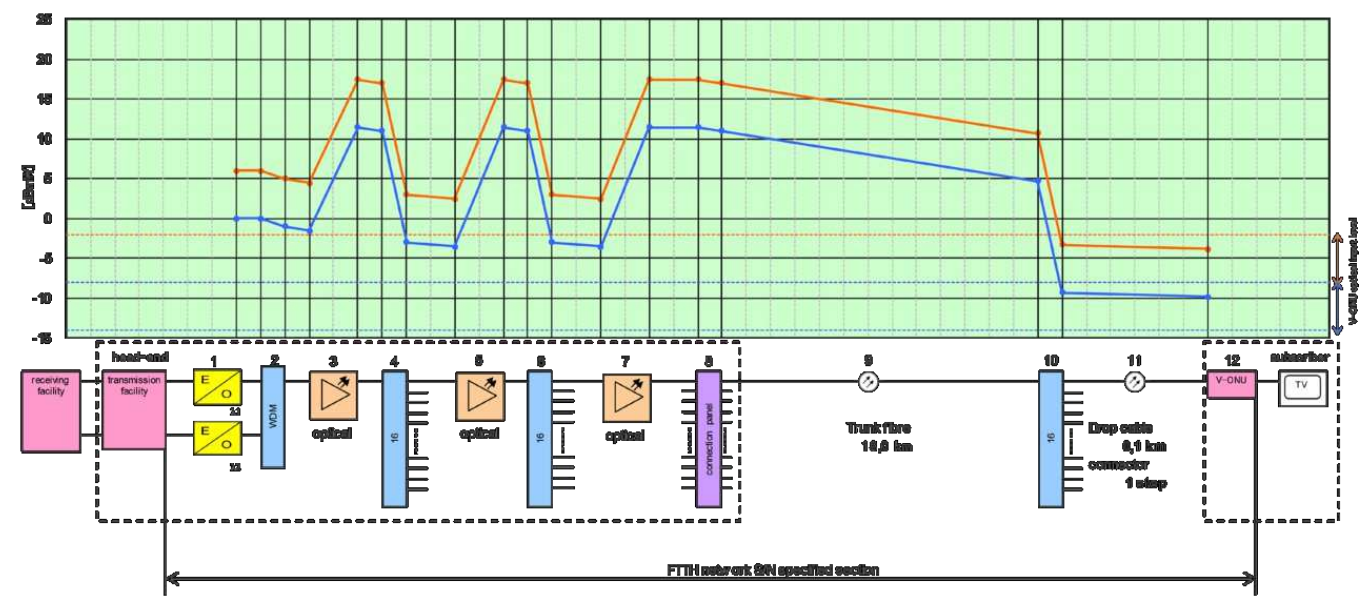


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Figure A.8 – System performance calculation for model C

Model D

Municipal type CATV, 3 EDFA stages, OMI: OFDM: 3,2 %, satellite (BS/CS): 4,3%
external and direct modulation optical transmitter: **2 wavelengths** (11 digital channels, 48 satellite (BS/CS) channels)



Item	
output RF	6,2 dB
OMI	3,2 % 4,3 %
RF input level	27,0 dB 10,0 dB

equipment	1	2	3	4	5	6	7	8	9	10	11	12
input level [dBmV]	RF:20dBmV	6,80	4,80	17,00	2,80	17,00	2,80	17,00	17,00	10,70	-3,20	-3,84
input level [dBmV]	RF:70dBmV	0,80	-1,80	11,00	-3,80	11,00	-3,80	11,00	11,00	4,70	-8,20	-8,84
gain or loss [dB]	-	-1,00	13,00	-14,00	18,00	-14,00	18,00	-14,00	-8,20	-14,00	-0,54	-
output level [dBmV]	6,80	5,80	17,80	3,80	17,80	3,80	17,80	17,00	10,70	-3,20	-8,84	RF:20dBmV
output level [dBmV]	6,80	-1,00	11,80	-3,00	11,80	-3,00	11,80	11,00	4,70	-8,20	-8,84	-
output RF [dBm] to 1)	-105,16	-	-182,73	-	-148,49	-	-147,82	-	-	-	-	-
output RF [dBm] to 1)	-103,60	-	-180,28	-	-145,87	-	-145,20	-	-	-	-	-
channel 201 [dB] note1	-	-	-	-	-	-	-	-	-	-	-	43,44
channel 202 [dB] note1	-	-	-	-	-	-	-	-	-	-	-	38,88

note 1: FTTH network ON at the V-ONU RF output, which is calculated from RF of the optical transmitter and the amplifier. (RF degradation when A1 and A2 are resolved is considered.)
note 2: 201(CF20240A4-2001 ,BNC2002) at V-ONU RF output, which is calculated from FTTH network ON
note 3: Gain 02 depending on input power and wave length of the optical amplifier is NOT combined in the above calculation.

Figure A.9 – System performance calculation for model D

Model E

Poor signal reception type CATV, without EDFA, OMI: OFDM: 6,4%, satellite (BS): 4,7%
directly modulated optical transmitter: **1 wavelength** (9 digital channels, 24 satellite (BS) channels)

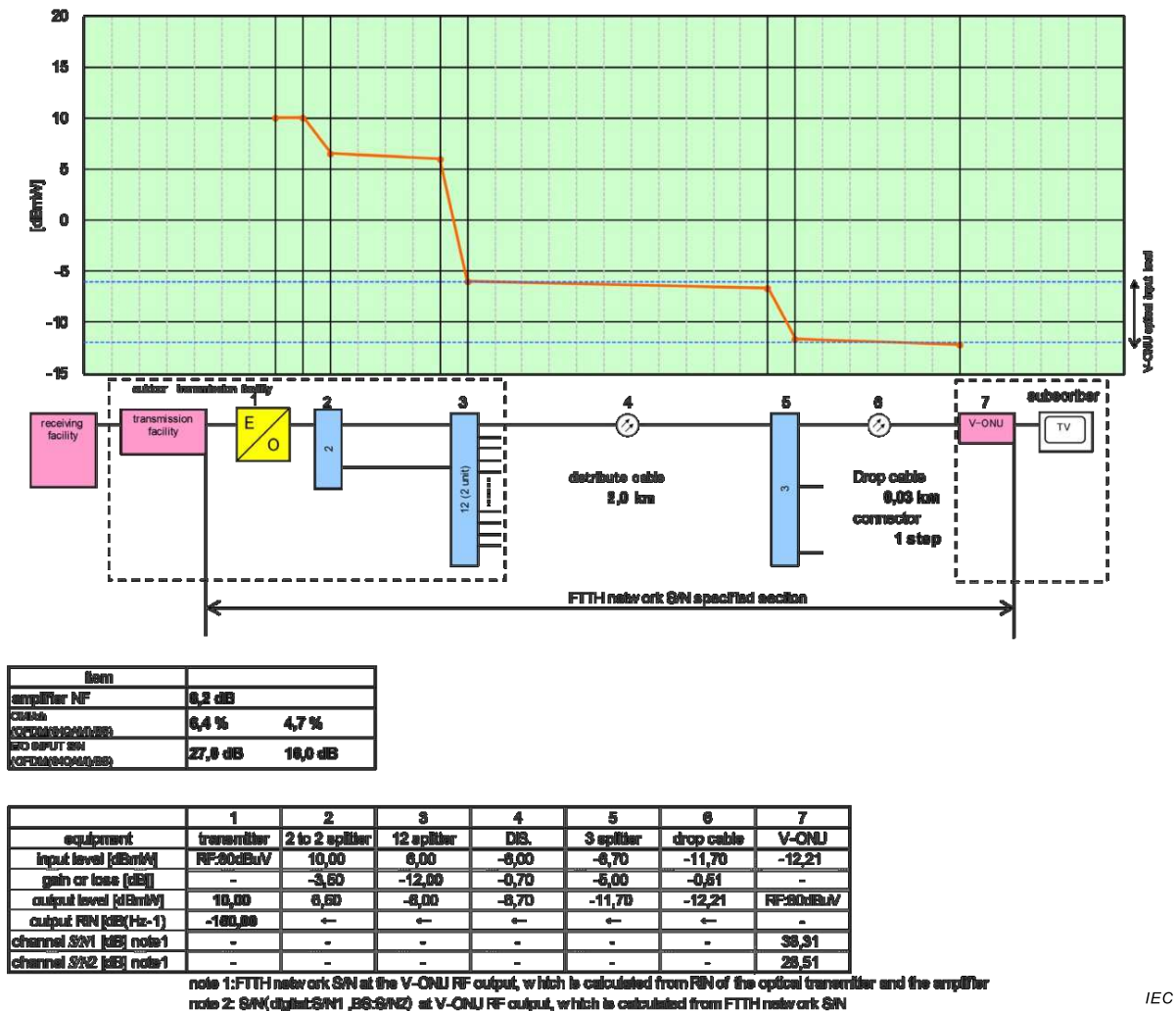


Figure A.10 – System performance calculation for model E

Model F

Poor signal reception type CATV, without EDFA, OMI: digital 6,4%, satellite (BS): 6,1%
directly modulated optical transmitters: **2 wavelengths** (9 digital channels, 24 satellite (BS) channels)

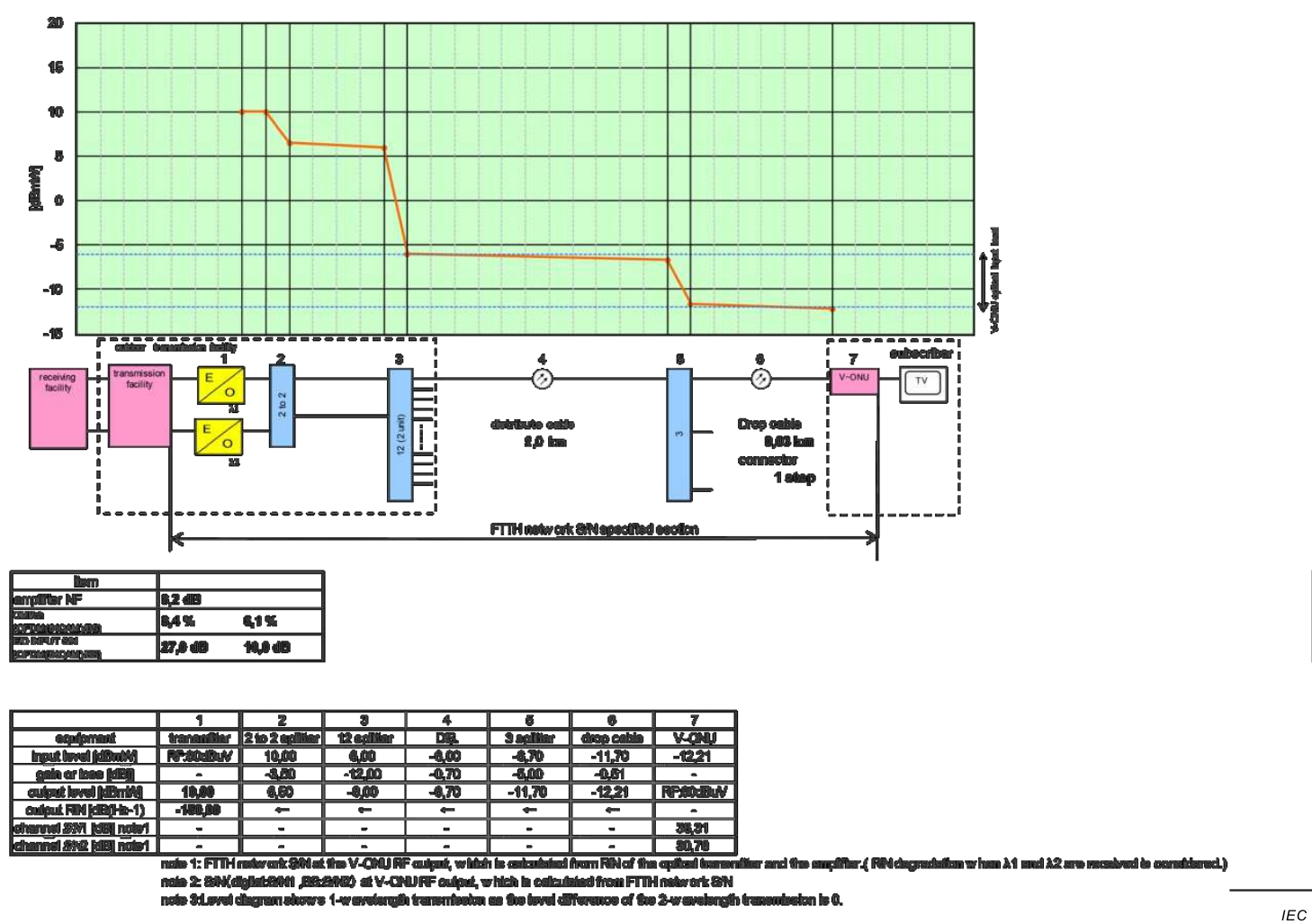


Figure A.11 – System performance calculation for model F

A.6 Guidelines for actual operation

A.6.1 Optical transmitter

It is considered to operate with the CS, BS and prospective satellite supplementary service facilities to control the SBS (stimulated Brillouin scattering) effect. For that purpose a directly modulated transmitter having a dummy carrier in the vicinity of 2 GHz is used.

A.6.2 Optical amplifier

In case of feeding multiple optical wavelengths (i.e. WDM system) to an amplifier, the gain tilt effect is considered. In general, the wider the wavelength spacing, the larger the gain tilt, and the operation levels become different. Careful attention should be paid to the level differences between wavelengths in cascaded operation. A 6 dB level difference is assumed between two wavelengths at the headend in this system, however, the gain tilt may impair the level difference and finally it becomes difficult to maintain the original level difference. This may degrade the S/N value.

Annex B (informative)

Wavelength division multiplexing

B.1 Optical wavelength grid (optical frequency grid)

The wavelength spacing is defined as follows.

a) DWDM frequency grid (ITU-T G.694.1):

There is a variety of channel spacings ranging from 12,5 GHz to 100 GHz and wider.

b) CWDM wavelength grid (ITU-T G.694.2):

A wavelength spacing of 20 nm is defined.

B.2 Nominal central frequencies and wavelengths

The wavelengths (frequencies) are defined as follows by two methods (ITU-T G.694.1):

a) DWDM case

For a channel spacing of 12,5 GHz, the allowed channel frequencies (in THz) are defined by:

$$f = 193,1 + n \times 0,012\,5 \text{ where } n \text{ is a positive or negative integer including } 0$$

For a channel spacing of 25 GHz, the allowed channel frequencies (in THz) are defined by:

$$f = 193,1 + n \times 0,025\,0 \text{ where } n \text{ is a positive or negative integer including } 0$$

For a channel spacing of 50 GHz on a fibre, the allowed channel frequencies (in THz) are defined by:

$$f = 193,1 + n \times 0,050\,0 \text{ where } n \text{ is a positive or negative integer including } 0$$

For a channel spacing of 100 GHz or more on a fibre, the allowed channel frequencies (in THz) are defined by:

$$f = 193,1 + n \times 0,100\,0 \text{ where } n \text{ is a positive or negative integer including } 0$$

The wavelength can be calculated by the following equation:

$$\lambda = \frac{c}{f}$$

where

λ	is the wavelength
f	is the frequency
c	is the speed of light in vacuum (= 2,997 924 58 × 10 ⁸ m/s)

The resulting approximate nominal central wavelengths are shown in Table B.1.

Table B.1 – Example nominal central frequencies of the DWDM grid

Approximate nominal central wavelengths nm	Nominal central frequencies in THz for spacings of	
	100 GHz	200 GHz
1 624,89	184,5	184,5
1 624,01	184,6	
1 623,13	184,7	184,7
1 622,25	184,8	
1 621,38	184,9	184,9
1 620,50	185,0	
1 619,62	185,1	185,1
1 618,75	185,2	
1 617,88	185,3	185,3
1 617,00	185,4	
1 616,13	185,5	185,5
1 615,26	185,6	
1 614,39	185,7	185,7
1 613,52	185,8	
1 612,65	185,9	185,9
1 611,79	186,0	
1 610,92	186,1	186,1
1 610,06	186,2	
1 609,19	186,3	186,3
1 608,33	186,4	
1 607,47	186,5	186,5
1 606,60	186,6	
1 605,74	186,7	186,7
1 604,88	186,8	
1 504,03	186,9	186,9
1 603,17	187,0	
1 602,31	187,1	187,1
1 601,46	187,2	
1 600,60	187,3	187,3
1 599,75	187,4	
1 598,89	187,5	187,5
1 598,04	187,6	
1 597,19	187,7	187,7
1 596,34	187,8	
1 595,49	187,9	187,9
1 594,64	188,0	
1 593,79	188,1	188,1
1 592,95	188,2	
1 592,10	188,3	188,3
1 591,26	188,4	

Approximate nominal central wavelengths nm	Nominal central frequencies in THz for spacings of	
	100 GHz	200 GHz
1 575,37	190,3	190,3
1 574,54	190,4	
1 573,71	190,5	190,5
1 572,89	190,6	
1 572,06	190,7	190,7
1 571,24	190,8	
1 570,42	190,9	190,9
1 569,59	191,0	
1 568,77	191,1	191,1
1 567,95	191,2	
1 567,13	191,3	191,3
1 566,31	191,4	
1 565,50	191,5	191,5
1 564,68	191,6	
1 563,86	191,7	191,7
1 563,05	191,8	
1 562,23	191,9	191,9
1 561,42	192,0	
1 560,61	192,1	192,1
1 559,79	192,2	
1 558,98	192,3	192,3
1 558,17	192,4	
1 557,36	192,5	192,5
1 556,55	192,6	
1 555,75	192,7	192,7
1 554,94	192,8	
1 554,13	192,9	192,9
1 553,33	193,0	
1 552,52	193,1	193,1
1 551,72	193,2	
1 550,92	193,3	193,3
1 550,12	193,4	
1 549,32	193,5	193,5
1 548,51	193,6	
1 547,72	193,7	193,7
1 546,92	193,8	
1 546,12	193,9	193,9
1 545,32	194,0	
1 544,53	194,1	194,1
1 543,73	194,2	

Approximate nominal central wavelengths nm	Nominal central frequencies in THz for spacings of	
	100 GHz	200 GHz
1 590,41	188,5	188,5
1 589,57	188,6	
1 588,73	188,7	188,7
1 587,88	188,8	
1 587,04	188,9	188,9
1 586,20	189,0	
1 585,36	189,1	189,1
1 584,53	189,2	
1 583,69	189,3	189,3
1 582,85	189,4	
1 582,02	189,5	189,5
1 581,18	189,6	
1 580,35	189,7	189,7
1 579,52	189,8	
1 578,69	189,9	189,9
1 577,86	190,0	
1 577,03	190,1	190,1

Approximate nominal central wavelengths nm	Nominal central frequencies in THz for spacings of	
	100 GHz	200 GHz
1 542,94	194,3	194,3
1 542,14	194,4	
1 541,35	194,5	194,5
1 540,56	194,6	
1 539,77	194,7	194,7
1 538,98	194,8	
1 538,19	194,9	194,9
1 537,40	195,0	
1 536,61	195,1	195,1
1 535,82	195,2	
1 535,04	195,3	195,3'
1 534,25	195,4	
1 533,47	195,5	195,5
1 532,68	195,6	
1 531,90	195,7	195,7
1 531,12	195,8	
1 530,33	195,9	195,9

b) CWDM case

The wavelengths to be used are defined in ITU-T G.694.2 (see Table B.2)

Table B.2 – Nominal central wavelength for spacing of 20 nm (ITU-T G.694.2)

Wavelength of laser diode nm	Nominal central wavelengths for spacing of 20 nm nm	Channel spacing nm
1 530 band	1 531	1 524,5 to 1 537,5
1 550 band	1 551	1 544,5 to 1 557,5
1 570 band	1 571	1 564,5 to 1 577,5
1 590 band	1 591	1 584,5 to 1 597,5
1 610 band	1 611	1 604,5 to 1 617,5

B.3 Notes regarding wavelength division multiplexing

B.3.1 Crosstalk between two wavelengths

When multiple wavelength signals are transmitted in a single fibre, crosstalk is generated. It is classified into linear and non-linear type.

a) Linear cross talk

Linear crosstalk is caused by the insufficient wavelength separation of the WDM filter. The residual frequency component of other wavelength signals appear at the receiving side due to the insufficient isolation (see Figure B.1).

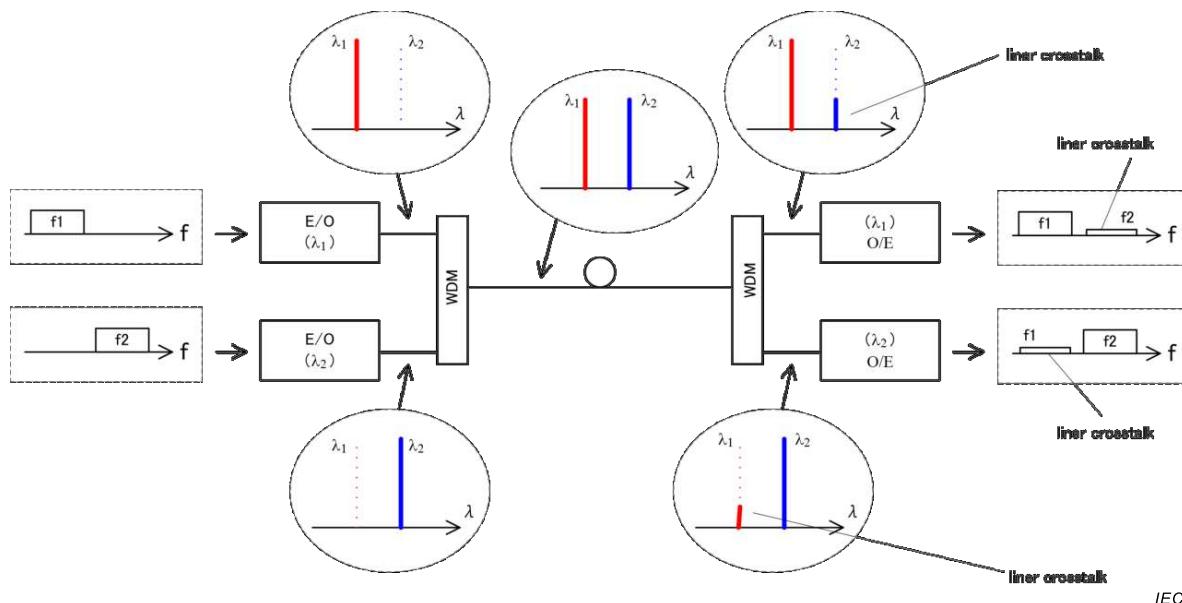


Figure B.1 – Linear crosstalk between two wavelengths

b) Non-linear cross talk

The typical non-linear crosstalk is Raman crosstalk which appears in the fibre itself. SRS (stimulated Raman scattering) is the cause of this Raman crosstalk. Raman crosstalk is the worst when the wavelength spacing is 13 THz (100 nm) as shown in Figure B.2. It is not possible to remove the residual frequency component of other wavelengths by the wavelength separation filter as shown in Figure B.3. XPM (cross-phase modulation) is another cause of non-linear crosstalk between different wavelength signals. The interference level may become significant in the case of high frequency usage (or small $\Delta\lambda$ wavelength separation) as shown in Figure B.4. However it is generally negligibly small for the FTTH system described in this document where different frequencies below and above 1 GHz are used.

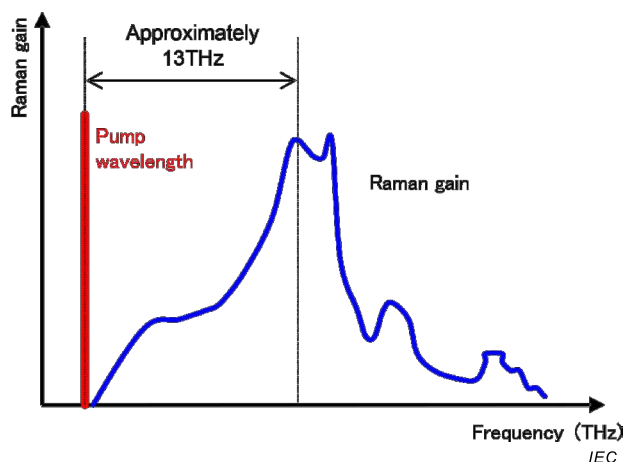


Figure B.2 – Wavelength dependency of Raman crosstalk

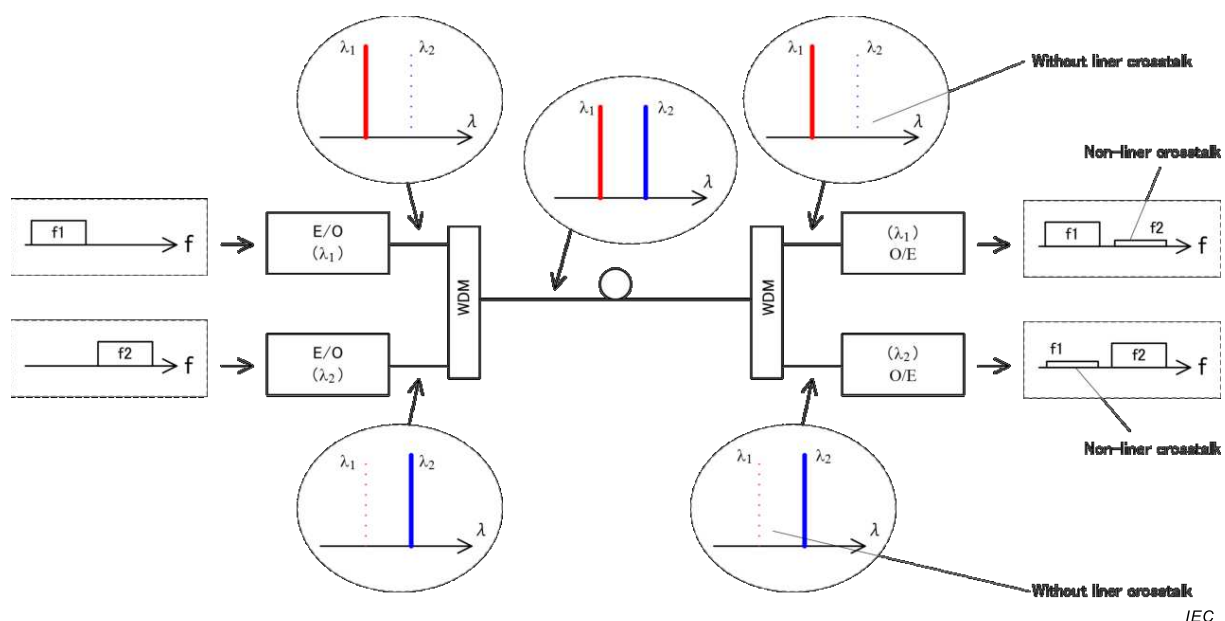


Figure B.3 – Nonlinear crosstalk between two wavelengths

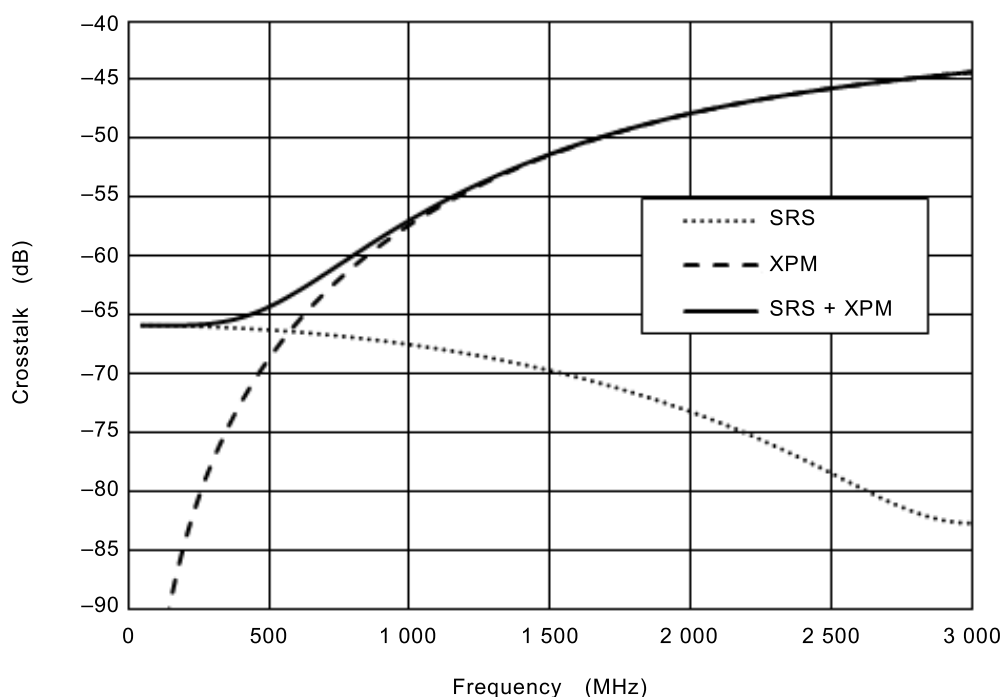


Figure B.4 – Frequency dependency of cross-phase modulation

(Under the condition of wavelength $\lambda_1 = 1\,555,00\text{ nm}$, $\lambda_2 = 1\,554,00\text{ nm}$, fibre length = 20 km and optical input power to fibre = 10 dB(mW))

B.3.2 Receiving two wavelengths by single V-ONU

When optical signals on the different two wavelengths are received by single V-ONU, the level difference of two RF signals (ΔC in dB) can be described as follows:

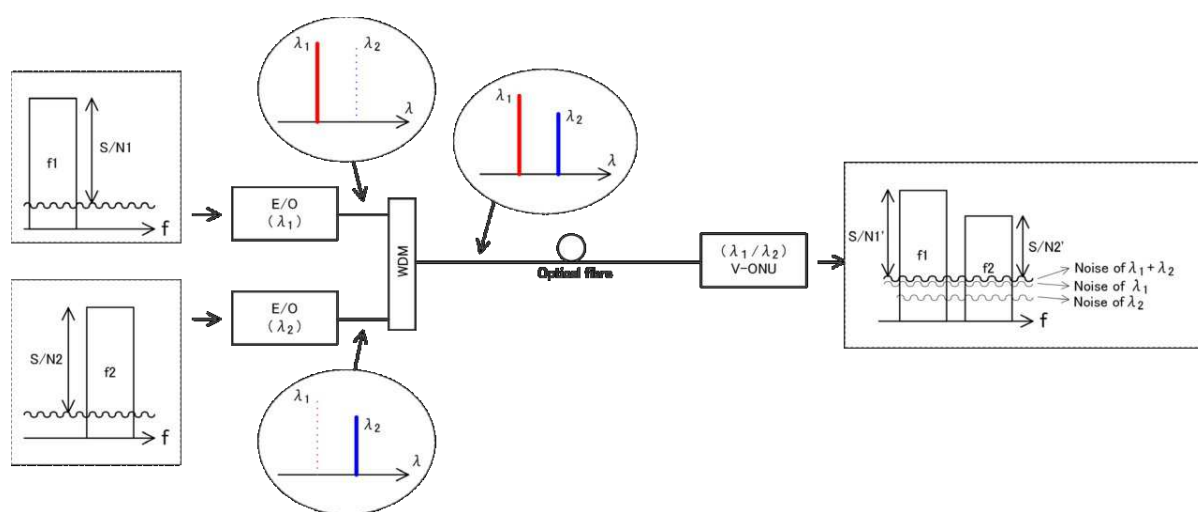
$$\Delta C = 2 \cdot 10 \cdot \lg\left(\frac{P_2}{P_1}\right) + 20 \cdot \lg\left(\frac{OMI_2}{OMI_1}\right) \text{ in dB} \quad (\text{B.1})$$

where

OMI_k is the optical modulation index of the k^{th} wavelength

P_k is the optical input level of the k^{th} wavelength

Figure B.5 shows the S/N degradation when optical signals on two wavelengths are received by one V-ONU. The noise of each optical carrier is accumulated and additional noise is generated by mutual interference between λ_1 and λ_2 . The calculation method of the total S/N is described in 6.5.5.



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Figure B.5 – S/N degradation (two wavelengths into one V-ONU case)

Annex C (informative)

Minimum wavelength separation

C.1 Optical beat interference

When signals of two wavelengths are received by a single optical receiver, an optical beat interference component called beat noise ($RIN_{\Delta\lambda}$) will be generated at a frequency centered around the difference of both wavelengths (frequencies). The centre frequency, f_{beat} of the optical beat noise is given by Equation (C.1).

$$f_{\text{beat}} = \frac{c|\lambda_i - \lambda_{i+1}|}{\lambda_i \cdot \lambda_{i+1}} \quad (\text{C.1})$$

where

c is the velocity of light in vacuum, $2,997\,924\,58 \times 10^8$ m/s

Also, the optical beat noise ($RIN_{\Delta\lambda}$) can be expressed using Equation (C.2).

$$RIN_{\Delta\lambda} = 10 \lg \left\{ \frac{2}{\pi} \cdot \frac{\Delta\nu}{(\Delta F - f)^2 + \Delta\nu} \right\} \text{ in dB} \quad (\text{C.2})$$

where

ΔF is the frequency equivalent of the wavelength difference

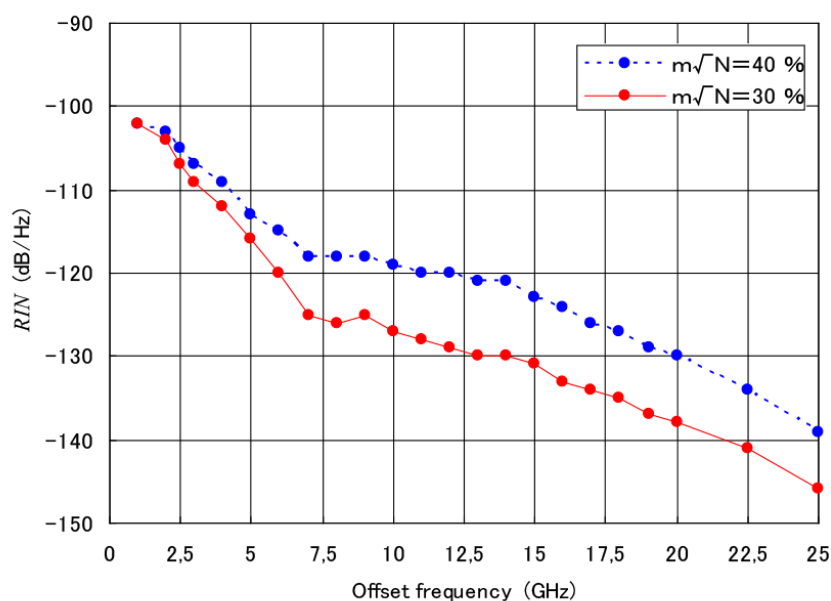
$\Delta\nu$ is the spectral width of the optical signal

f is any frequency for RIN calculation

In Figure C.1 experimental results of RIN degradation due to optical beat are plotted against the offset frequency (difference between centre frequency of optical beat and RF frequency transmitted through the optical signal). For example, if the optical signal is modulated by an RF frequency of 2,5 GHz (assuming the total optical modulation index is 30 %), and if the required RIN is $-140 \text{ dB}(\text{Hz}^{-1})$, then the centre frequency of beat noise should be

$$f_{\text{beat}} \geq 25 \text{ GHz} \quad (\text{C.3})$$

25 GHz corresponds to 0,20 nm in the 1 550 nm band. Therefore, it is sufficient that the two wavelengths are separated by at least 0,20 nm to avoid the effect of noise due to optical beat interference. However, in a large scale FTTH system such as satellite signal transmission, the required RIN value is less than $-150 \text{ dB}(\text{Hz}^{-1})$. In this case, the minimum required wavelength separation including allowance is 0,3 nm. In actual DWDM wavelengths systems (like ITU-T 100 GHz grid), the wavelength separation will be approximately 0,8 nm and there is no need to consider the optical beat interference. Similarly, in case of CWDM wavelength systems, the wavelength separation will be 20 nm and again there is no need to take into account the optical beat interference.



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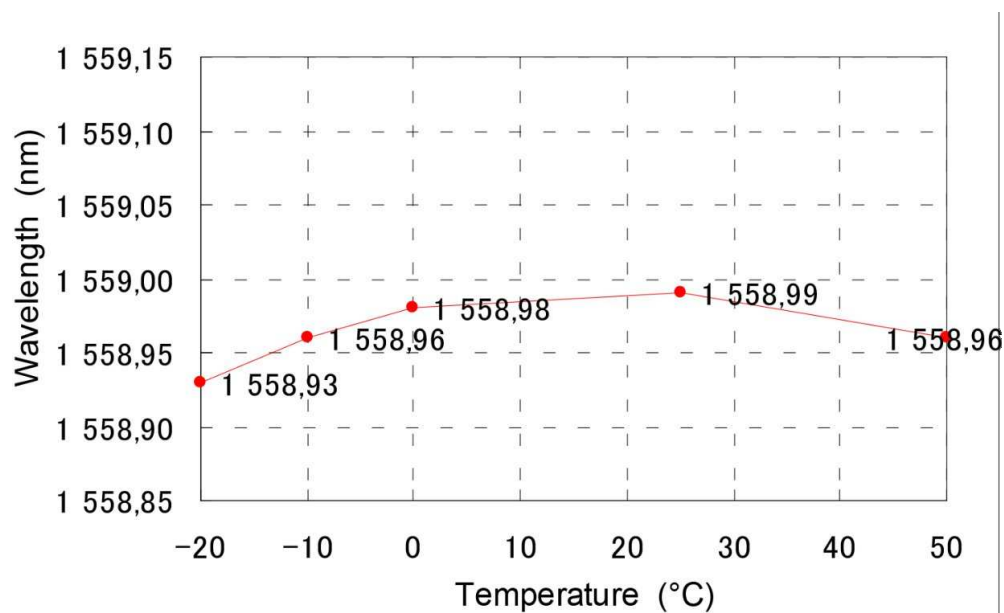
Figure C.1 – Experimental results of *RIN* degradation due to optical beat

Other types of optical WDM systems may contain transmitters with automatic temperature control (ATC) function. A 100 GHz ITU-T grid system requires a minimum wavelength separation between the multiplexed optical signals of at least 0,8 nm. In order to meet this requirement the assumed wavelength variation is 0,2 nm for ATC circuit, 0,3 nm for long-term usage and 0,3 nm for general tolerance in actual systems.

C.2 Range of wavelength variation

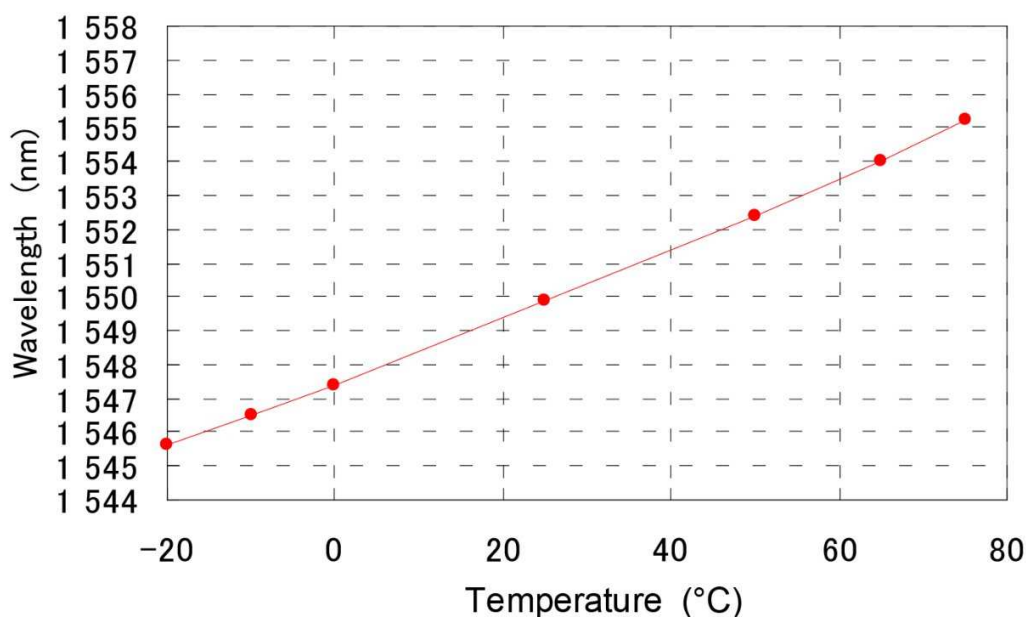
Optical transmitters may contain ATC circuits to control the variation of optical wavelength against temperature. DWDM transmitters usually have an ATC to minimize the wavelength variation. However, typical CWDM transmitters do not employ ATC, as a result the wavelength may shift by 0,1 nm to the longer (or shorter) wavelength side if the temperature increases (or decreases) by 1 °C.

Experimental results on the variation of the optical wavelength of a DWDM transmitter are shown in Figure C.2, and for a CWDM transmitter, in Figure C.3.



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Figure C.2 – Wavelength variation of a DWDM transmitter against ambient temperature

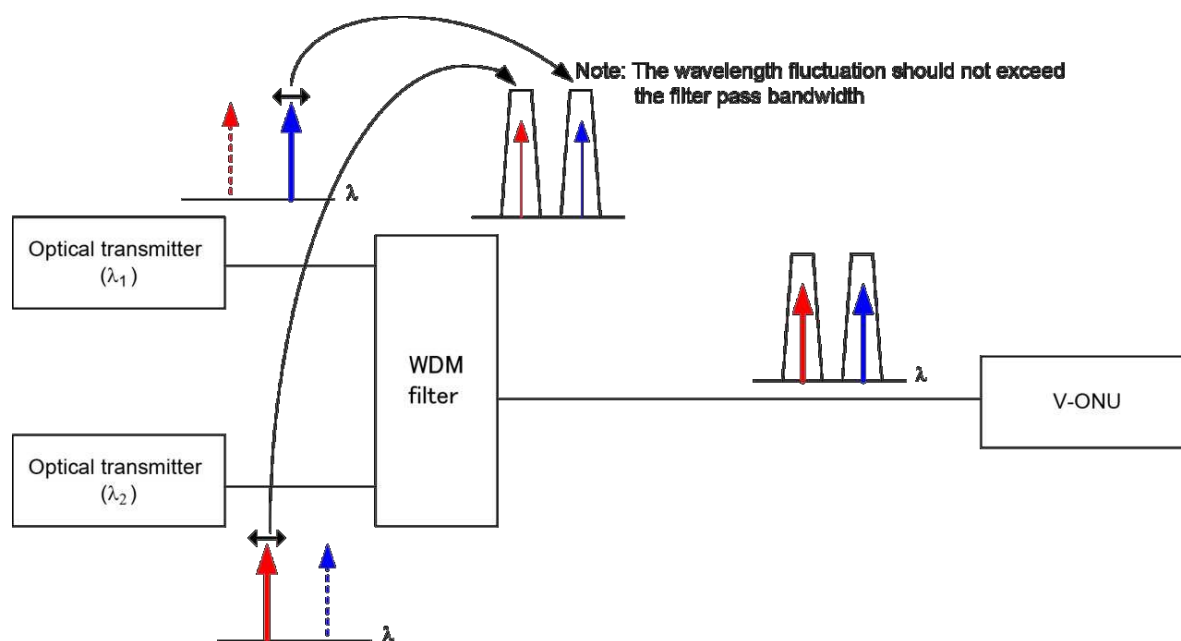


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Figure C.3 – Wavelength variation of a CWDM transmitter against ambient temperature

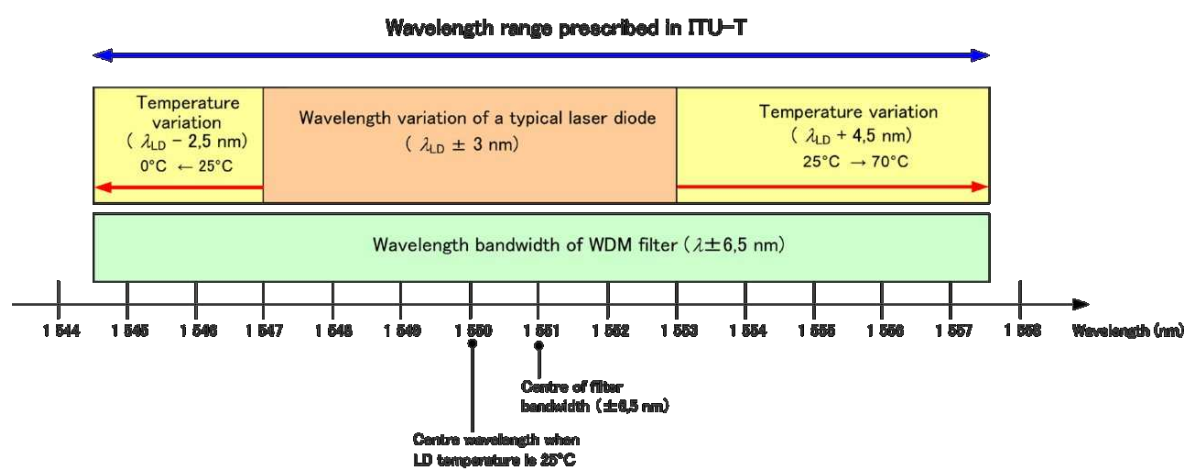
C.3 WDM system using optical filters and couplers

WDM systems can be designed with WDM filter or optical coupler. For WDM systems with WDM filter, care should be taken that the wavelength variation does not exceed the pass bandwidth of the WDM filter, as illustrated in Figure C.4. Figure C.5 shows a CWDM filter design taking into account temperature and LD wavelength variation. Figure C.6 shows an example of a wavelength division multiplex system using an optical coupler. It is noted that the wavelength separation shall be kept in at least 0,3 nm.



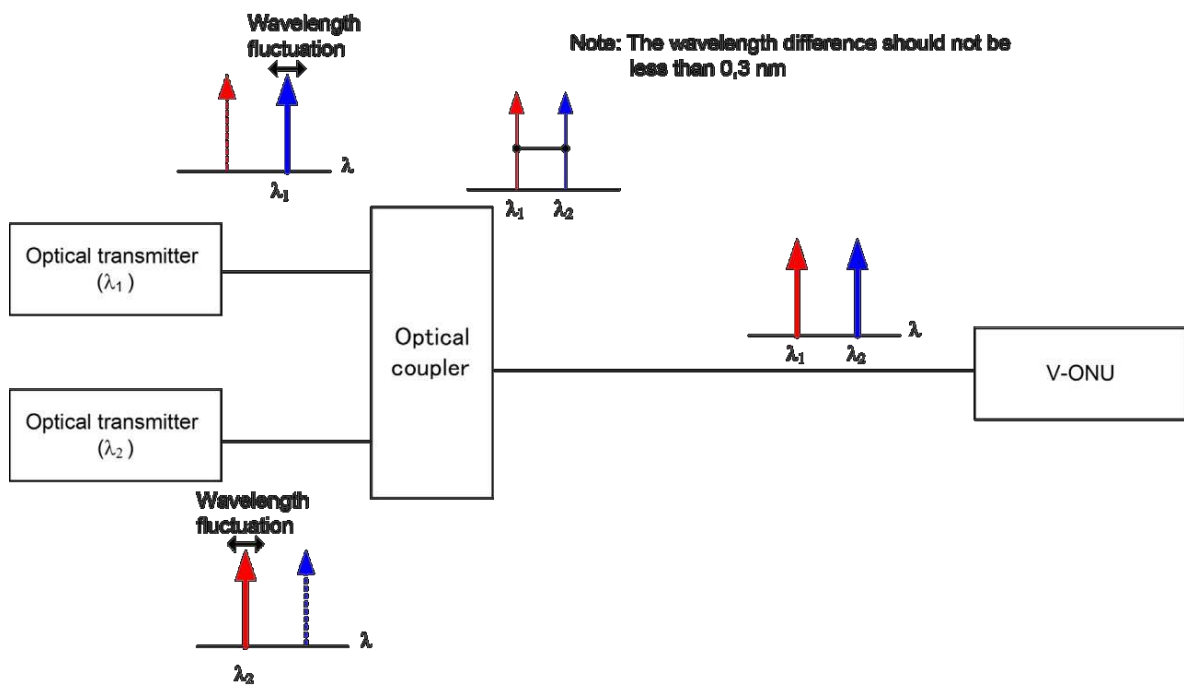
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Figure C.4 – Example of wavelength division multiplexing using WDM filter



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Figure C.5 – Example of CWDM filter design



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Figure C.6 – Example of wavelength division multiplexing using optical coupler

Annex D (informative)

Relation between S/N degradation and rain attenuation

The ratio of antenna aerial gain to the noise temperature of a receiving system, G/T is related to S/N by the following expression:

$$G/T = \left[\frac{S/N \cdot K \cdot B}{EIRP \cdot L_S \cdot R \cdot N_U} \right] \quad (D.1)$$

S/N is given by

$$S/N = \left[\frac{G/T \cdot EIRP \cdot L_S \cdot R \cdot N_U}{K \cdot B} \right] \quad (D.2)$$

where

- K is Boltzmann's constant
- B is the channel bandwidth
- $EIRP$ is the equivalent isotropically radiated power
- L_S is the propagation loss in the free space
- R is the rain attenuation
- N_U is the receiving S/N degradation due to uplink S/N

G/T can be expressed by the following expression.

$$G/T = \left[\frac{\alpha \cdot \beta \cdot G_r}{\alpha \cdot T_a + (1 - \alpha) \cdot T_0 + (F - 1) \cdot T_0} \right] = \left[\frac{\alpha \cdot \beta \cdot G_r}{\alpha \cdot T_a + (F - \alpha) \cdot T_0} \right] \quad (D.3)$$

where

- α is the coupling loss
- β is the pointing loss
- G_r is the gain of the receiving antenna
- T_a is the noise temperature of the receiving antenna
- T_0 is the reference temperature
- F is the noise figure of the receiving system

S/N degradation due to rain attenuation is defined as the difference between S/N during clear weather and S/N during rainy weather.

$$S/N \text{ degradation} = S/N_1 - S/N_2 =$$

$$10 \lg \left[\frac{G/T_1 \cdot EIRP \cdot L_S \cdot N_U}{K \cdot B} \right] - 10 \lg \left[\frac{G/T_2 \cdot EIRP \cdot L_S \cdot R \cdot N_U}{K \cdot B} \right] =$$

$$10 \lg[G/T_1] - 10 \lg[G/T_2] - 10 \lg[R] =$$

$$10 \lg \left[\frac{\alpha \cdot T_{a2} + (F - \alpha) \cdot T_0}{\alpha \cdot T_{a1} + (F - \alpha) \cdot T_0} \right] - 10 \lg [R] [\text{dB}] \quad (\text{D.4})$$

where

S/N_1 is the S/N value during clear weather

S/N_2 is the S/N value during rainy weather

G/T_1 is the G/T value during clear weather

G/T_2 is the G/T value during rainy weather

Assuming the noise temperature, T_{a1} of the receiving antenna during clear weather to be 50 K, the noise temperature, T_{a2} of the receiving antenna during rainy weather can be expressed by the following relation.

$$T_{a2} = 50 \cdot R + (1 - R) \cdot T_0 \quad (\text{D.5})$$

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